



Bayesian registration of functions with compositional and additive noises

Woo Min Kim^{a,*}, Yijia Ma^b, Wei Wu^a

^a Department of Statistics, Florida State University, Tallahassee, FL, 32306, USA

^b Department of Psychiatry and Behavioral Sciences, University of Texas Health Science Center, Houston, TX, 77030, USA

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ABSTRACT

Function registration (or alignment) is a fundamental challenge in functional data analysis, where the goal is to identify compositional variability and remove such “noise” before conducting in-depth analysis such as component methods and regressions. While significant progress has been made in this area over the past few decades, the registration of observations with both compositional and additive noises remains understudied. Most state-of-the-art methods, including the well-known Fisher-Rao approach, rely on derivatives and often cannot adequately account for additive noise. A Bayesian framework is proposed that overcomes these limitations by directly modeling likelihoods in the original function space, enhancing robustness to additive noise. Moreover, the centered log-ratio transformation is employed to represent time-warping functions, mapping them to a square-integrable subspace that is well-suited for stochastic process priors. The framework is extended to conduct signal estimation in the presence of compositional noise, additive noise, constant translation, and constant scaling, achieving desirable results. Through extensive simulations and real-data applications, the superior performance and robustness of the new method over state-of-the-art approaches are demonstrated.

1. Introduction

Function registration is a central problem in functional data analysis. Functional observations typically exhibit two primary sources of variability: *amplitude* and *phase*. Given a time domain $t \in \mathcal{I}$, where we assume without loss of generality that $\mathcal{I} = [0, 1]$, a function f is defined on $\mathbb{L}^2([0, 1])$ space. Amplitude variability refers to differences in the values of $f(t)$ across functions at fixed time points. In contrast, phase variability reflects differences in the timing or location of salient features such as peaks and valleys. Ignoring phase variability and treating misaligned functions as if already synchronized can lead to misleading statistical inferences, a concern illustrated in the classical textbook on functional data analysis (Ramsay and Silverman, 2006), where averaging phase-shifted curves distorts the meaningful structure.

Phase variation is typically modeled by time warping functions, denoted by γ . These functions belong to the set

$$\Gamma \triangleq \{ \gamma : [0, 1] \rightarrow [0, 1] | \gamma(0) = 0, \gamma(1) = 1, 0 < \dot{\gamma}(t) < \infty \},$$

where the derivative function $\dot{\gamma}$ ensures strict monotonicity and finite, smooth transformations of time. A seminal contribution by Sakoe and Chiba (1978) applied dynamic programming to align speech signals, offering one of the earliest algorithmic solutions for time warping, although it did not enforce smoothness or a strictly increasing property in γ .

* Corresponding author.

E-mail address: wkim6@fsu.edu (W.M. Kim).

Early statistical approaches to registration focused on aligning the timing of specific landmarks, distinct structural features such as peaks or inflection points. Landmark-based registration was studied in depth by Kneip and Gasser (1992) and Gasser and Kneip (1995), who formalized concepts such as structural points and structural averaging. However, this method faces limitations: landmarks may be ambiguous or noisy in certain curves, and aligning only a few discrete points may fail to correct variability in other regions. Moreover, landmark identification is typically manual, making it impractical for large-scale datasets.

To address these limitations, subsequent methods sought global alignment by minimizing a least-squares criterion over the entire function domain. Given observed functions $f_1, \dots, f_N \in \mathbb{L}^2([0, 1])$, each alignment problem can be formulated as:

$$\hat{\gamma}_i = \arg \inf_{\gamma \in \Gamma} \|g - (f_i \circ \gamma)\|_{\mathbb{L}^2}^2 + \lambda \mathcal{R}(\gamma),$$

where $g \in \mathbb{L}^2([0, 1])$ is a unknown template function (to which all f_i are aligned), $\mathcal{R}(\gamma)$ is a regularization term encouraging smoothness, and λ controls its strength. We denote $\|\cdot\|_{\mathbb{L}^2}$ as the $\mathbb{L}^2$ metric. In practice, γ is often modeled using B-spline basis expansions, enabling flexible yet smooth monotonic transformations. This approach remains widely used, as seen in Gervini and Gasser (2004), James (2007), and Eilers (2004).

A recent and highly effective approach for functional registration is rooted in the *square-root velocity function* (SRVF) representation (Srivastava et al., 2011). This framework leverages the fact that the $\mathbb{L}^2$ distance between SRVFs corresponds to a special case of the Fisher-Rao metric (Rao, 1945), a unique Riemannian metric invariant under time warping reparametrization (Čencov, 1982). This invariance is highly beneficial because it ensures that the metric comparison of functional data remains consistent regardless of arbitrary time warping, allowing for a robust and meaningful assessment. Moreover, directly computing the Fisher-Rao metric is often challenging (Bhattacharya, 1943); the SRVF formulation significantly simplifies this computation. In practice, functional registration is achieved by optimizing a least-squares criterion in SRVF space to estimate optimal time warping functions, typically implemented via dynamic programming.

Previous studies have demonstrated that the optimization-based Fisher-Rao (F-R) framework is effective in identifying optimal time warping functions. However, the optimization procedure produces only a single (global) optimum and inherits two practical caveats: (1) it does not quantify uncertainty in the alignment, and (2) it may be sensitive to additive noise. Both issues become particularly important in real-world applications, where ground-truth warpings are unobservable and observed data is often noisy. Regarding (1), the lack of uncertainty quantification is problematic when multiple reasonable warpings exist. For instance, if one function has two prominent peaks and the other has three, there may be multiple reasonable ways to align them depending on which peaks are synchronized, leading to distinct yet plausible warpings. As for (2), the method's sensitivity to noise arises from its reliance on the SRVF, a function of the first derivative of the observed signal. These two issues will be elaborated upon in Section 3.

As a probabilistic model-based approach, the Bayesian framework addresses these shortcomings by treating the warping functions as a stochastic process. It can represent multiple most probable alignments along with credible intervals, which reflect the uncertainty in the estimated warpings. Furthermore, Bayesian methods naturally accommodate prior information, enabling more flexible and robust modeling of time warping. Several Bayesian methods have been developed within the SRVF framework. While these methods share a common foundation in the SRVF framework, they differ in the representation of the warping functions, the choice of prior distributions, and the inference techniques for posterior sampling. Cheng et al. (2016) placed a Dirichlet prior on the increments of discretized warpings, which are then interpolated to form valid reparameterizations. To overcome limitations associated with discretization and to more naturally incorporate prior information, Kurtsek (2017) introduced a prior on warping functions using smooth orthonormal basis functions defined in the tangent space of the square-root density (SRD) manifold, and performed posterior inference using importance sampling defined over the manifold of SRVFs of warping functions. Building on this geometric framework, Lu et al. (2017) proposed a Gaussian process (GP) prior on the SRD tangent space and employed the preconditioned Crank-Nicolson (pCN) MCMC method for sampling. Tucker et al. (2021) adopted the same GP prior structure but improved sampling efficiency by using an infinite-dimensional Hamiltonian Monte Carlo (∞ -HMC) approach. More recently, Ma and Wu (2025) proposed a Bayesian model that applies the *centered log-ratio* (CLR) (Egozcue et al., 2006) transformation to warping functions, mapping them into an $\mathbb{L}^2$ subspace using an isometric isomorphism where Gaussian process priors can be placed directly. This approach allows for smooth inference while naturally enforcing positivity, integrability, and boundary constraints.

Although this paper focuses on functional registration, it is closely related to the broader literature on functional regression and functional data modeling in other domains, including spatial statistics, spatio-temporal modeling, and imaging data analysis. In spatial and spatio-temporal statistics, stochastic-process and Bayesian hierarchical models provide a general framework for modeling spatially and temporally indexed data (Cressie and Wikle, 2011). Related Bayesian work has developed scalable inference for GP-based varying-coefficient models and massive spatial data (Guhaniyogi et al., 2022, 2023). Functional regression models have also been widely used in imaging-data applications (Morris, 2015). These approaches model functional, spatial, or imaging data using rich regression and stochastic-process structures, but they do not primarily address registration uncertainty arising from subject-specific phase variation. Our framework is complementary to this literature: it focuses on estimating monotone warping functions and quantifying alignment uncertainty for noisy functional observations.

In this paper, we propose a new Bayesian registration framework for functional observations with both compositional variability and additive noise. In particular, for observations $y_1, \dots, y_N$ generated from the following model:

$$y_i(t) = c_i g \gamma_i^{-1}(t) + e_i + \epsilon_i(t), \text{ and } i = 1, \dots, N. \quad (1)$$

The goal is to estimate the underlying signal $g \in \mathbb{L}^2([0, 1])$ from the observed functions. Here, $\gamma_i \in \Gamma$ is a subject-specific monotone warping function that captures phase variation. The parameters $c_i \in \mathbb{R}$ and $e_i \in \mathbb{R}$ represent curve-specific global scale and translation effects, respectively. The error term $\epsilon_i \in \mathbb{L}^2([0, 1])$ represents additive observation noise in the original function space.

Table 1
Positioning relative to proximal prior art (ordered by year).

Method	Inference	Warping rep.	Objective/Likelihood domain	Main distinction vs. ours (related contrib.)
Telesca and Inoue (2008)	Bayesian	Monotone B-splines	Original space	Warplings are finite-dimensional (B-spline coefficients), not CLR-space GP regularization.
Srivastava et al. (2011)	Optimization	SRD	SRVF space	Point estimate only; SRVF involves differentiation, so high-frequency observation noise can be amplified.
Lu et al. (2017)	Bayesian	SRD	SRVF space	Likelihood is defined on SRVF space, not directly on $y(t)$ under Eq. (1)
Bharath and Kurtek (2020)	Bayesian	Warp maps (PL sampling)	SRVF space	SRVF-based pairwise alignment with a prior/proposal on warp maps; does not target our joint Eq. (1) model.
Ma and Wu (2025)	Bayesian	CLR	SRVF space	Uses CLR for warpings within an SRVF-likelihood setting
Proposed method	Bayesian	CLR	Original space	Original-space likelihood for Eq. (1) CLR-GP regularization; targets joint registration, signal estimation under additive noise.

To address model identifiability, we impose centering constraints on the scale, translation, and warping components. Specifically, we assume $\mathbb{E}[e_i] = 1$, $\mathbb{E}[e_j] = 0$, and that the warping functions are centered around the identity warping $\gamma_{\text{id}}(t) = t$ (the formal definition of this warping centering is given in [Section 2](#)). We also assume that the additive noise has zero mean, i.e., $\mathbb{E}[e_i(t)] = 0$ for any $t \in [0, 1]$.

When the additive noise is not present (i.e., $e_i(t) \equiv 0$), the signal estimation is well addressed by the F-R framework ([Kurtek et al., 2011](#)) and other Bayesian frameworks ([Cheng et al., 2016](#); [Lu et al., 2017](#); [Tucker et al., 2021](#); [Ma and Wu, 2025](#)), all of which rely on SRVFs of the original observations, and such an approach must be based on the smoothness assumption regarding the observations. A recent study by [Tucker et al. \(2021\)](#) examined pairwise alignment when both compositional noise and additive noise are present. As SRVF is still used in both likelihood and prior models, the procedure may involve undesired complexity and non-robustness.

Rather than aligning functions in their SRVF representation, we propose performing alignment directly in the original function space. The main procedure of our method is the Bayesian inference for estimating both the template function g and the CLR-transformed warping functions γ_i 's. We place Gaussian process priors on both components to reflect smoothness assumptions and sample the posterior draws using the Metropolis-Hastings algorithm within Gibbs sampling. For posterior sampling method, we adopt the Z-mixture pCN algorithm ([Lu et al., 2017](#)), a variant of the pCN algorithm originally proposed by [Cotter et al. \(2013\)](#). This Metropolis-Hastings method is particularly well-suited for infinite-dimensional function spaces and enables efficient sampling of random functions. The Z-mixture extension improves performance in the presence of multimodal posteriors by allowing occasional large jumps between modes. In contrast, the ∞ -HMC method used in [Tucker et al. \(2021\)](#) requires multiple MCMC chains to adequately explore multimodal posteriors and involves more intricate hyperparameter tuning, which could lead to inefficiency in practice.

Our contributions in this paper can be summarized in the following two points:

1. We propose a unified Bayesian framework for functional registration and signal estimation under the generative model in [Eq. \(1\)](#), which includes compositional warping variability, additive noise, and signal-specific global scale/shift effects.
2. We establish a robust alternative to the prevailing SRVF framework. We show that functional alignment can be carried out with the likelihood specified in the original observation space, even under substantial additive noise. Under our CLR-GP prior, the induced regularization mitigates pinching behavior in practice.

[Table 1](#) compares our framework with closely related methods. The main differences are the warping representation and the space in which the objective or likelihood is defined. For example, [Telesca and Inoue \(2008\)](#) used monotone B-spline warpings in an original-space Bayesian model, while [Srivastava et al. \(2011\)](#) used an SRVF-based optimization criterion. [Lu et al. \(2017\)](#) used an SRVF-based likelihood, and [Bharath and Kurtek \(2020\)](#) constructed distributions on warp maps that can be used as priors in Bayesian alignment. More recently, [Ma and Wu \(2025\)](#) used CLR warpings with an SRVF-based likelihood. In contrast, our framework combines an original-space likelihood under [Eq. \(1\)](#) with CLR-based Gaussian process regularization for monotone warpings under the noisy generative model.

The effectiveness of the new method will be demonstrated through various simulations and real-world datasets. Additionally, we will compare its performance with that of state-of-the-art registration methods.

The following part of this paper is organized as follows. We begin with a review of background concepts in [Section 2](#), covering the CLR transformation for time-warping functions in depth. We then introduce our proposed method in detail, focusing on functional registration that involves both pairwise and multiple functions, as well as signal estimation procedures under a data-generation model. Each case is illustrated using a simulation example. [Section 3](#) presents (1) more simulation studies that offer deeper analysis, highlighting the flexibility and robustness of our approach, respectively, and (2) real data experiments that demonstrate its practical effectiveness. In [Section 4](#), we discuss the advantages of our method compared to the F-R framework and other existing Bayesian approaches.

2. Bayesian registration

The registration method proposed in this paper builds on the Bayesian framework introduced by [Ma and Wu \(2025\)](#), which focused on functional registration using the CLR representation of time-warplings in the prior. However, on top of that, the new model is more effectively able to deal with the registration problem even when including the presence of both compositional and additive noises. By modeling warping functions using the CLR transformation and applying priors directly in the original function space, we depart from conventional SRVF-based methods. This not only enhances robustness and computational tractability but also naturally enforces boundary conditions and positivity constraints in the set Γ .

2.1. Review of the CLR prior in a Bayesian framework

Previous studies ([Lu et al., 2017](#); [Tucker et al., 2021](#)) have commonly adopted the SRVF representation for modeling time-warping functions, which is widely regarded as a state-of-the-art approach in functional data alignment. In this framework, each warping function γ is mapped to $\sqrt{\dot{\gamma}}$ (where the dot operation indicates derivative), which is in the positive orthant of the unit sphere in a nonlinear Hilbert sphere, referred to as the SRD (square-root density) space:

$$\mathbb{S}_{\infty}^{+} = \left\{ q_{\gamma} \in \mathbb{L}^2 \mid q_{\gamma}(t) \geq 0, \int_0^1 q_{\gamma}^2(t) dt = 1 \right\}.$$

Statistical modeling is typically conducted in the tangent space of the SRD manifold at a reference point p , commonly chosen as the SRVF of the identity warping $\gamma_{\text{id}}(t) = t$, for which $p(t) = 1$. The tangent space, $T_p(\mathbb{S}_{\infty}^{+})$, provides a local linear approximation to the nonlinear SRD manifold and enables the use of Gaussian process priors and other tractable statistical models.

However, the exponential map, $\exp_p : T_p(\mathbb{S}_{\infty}^{+}) \rightarrow \mathbb{S}_{\infty}^{+}$, is only locally bijective and does not preserve global structure. This approximation introduces geometric distortion when sampled points lie far from the reference, as the tangent space is not isometrically isomorphic to the manifold. This issue becomes more pronounced when modeling is restricted to the positive orthant $\mathbb{S}_{\infty}^{+}$, as required for time-warping functions. In Bayesian inference, Monte Carlo sampling in the SRVF framework must account for this constraint, necessitating an additional post-processing step to filter out or correct samples that fall outside the admissible space. This can introduce both computational inefficiency and potential bias in the resulting posterior inference.

To address these issues, [Ma et al. \(2024\)](#) proposed the use of the CLR transformation ([Egozcue et al., 2006](#)) as an alternative representation of warping functions. The CLR transformation maps warping functions into a linear, inner product space that is isometrically isomorphic to a subspace of the Euclidean $\mathbb{L}^2$ space, allowing for more flexible and principled statistical modeling.

More formally, we consider warping functions in Γ_1 , where

$$\Gamma_1 = \{ \gamma : [0, 1] \rightarrow [0, 1] \mid \gamma(0) = 0, \gamma(1) = 1, 0 < m_{\gamma} < \dot{\gamma}(t) < M_{\gamma} < \infty \}.$$

The lower bound m_{γ} and upper bound M_{γ} can guarantee finite moments, which are necessary for stochastic modeling purposes. The CLR transformation $\psi : \Gamma_1 \rightarrow H(0, 1)$ is defined by

$$H(0, 1) = \left\{ h \in \mathbb{L}^2([0, 1]) \mid \int_0^1 h(t) dt = 0, -\infty < m_h < h(t) < M_h < \infty \right\},$$

and provides an isometric isomorphism via:

$$h(t) = \psi(\gamma)(t) = \log(\dot{\gamma}(t)) - \int_0^1 \log(\dot{\gamma}(s)) ds, \text{ and } \gamma(t) = \psi^{-1}(h)(t) = \frac{\int_0^t \exp(h(s)) ds}{\int_0^1 \exp(h(s')) ds'}. \quad (2)$$

Unlike the SRVF representation, the CLR framework enables a more expressive and robust characterization of the warping space Γ , while avoiding strict positivity constraints. This makes it especially advantageous for Bayesian modeling, as it simplifies both the geometry and computation involved in functional inference.

[Fig. 1](#) illustrates the advantages of using the CLR transformation over the SRVF framework for modeling and resampling warping functions. Panel (a) shows 100 randomly generated warping functions, constructed using the first two orthonormal Fourier basis functions, i.e., $\{\phi_1(t), \phi_2(t)\} = \{\sqrt{2} \cos(2\pi t), \sqrt{2} \sin(2\pi t)\}$ for $t \in [0, 1]$. Each function is defined as $\gamma_i = \psi^{-1}(c_{1,i}\phi_1 + c_{2,i}\phi_2)$, where $c_{1,i} \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, 0.6^2)$ and $c_{2,i} \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, 0.4^2)$ for $i = 1, \dots, 100$. Given these observed warping functions, resampling is performed using the FPCA-based method described in [Ma et al. \(2024\)](#). Specifically, FPCA is performed on the observed warping functions either (1) in the tangent space of the SRD manifold for the SRVF-based model, or (2) in $H(0, 1)$ for the CLR-based model. In each case, resampled functions are generated by drawing principal component scores from zero-mean Gaussian distributions with variances estimated from FPCA, and reconstructing functions via basis expansion using the corresponding principal component functions in the respective representation space (either the tangent space of the SRD manifold or the CLR space). The resulting functions are then mapped back to the warping function space, employing the inverse SRVF transformation for the SRVF-based model, i.e., $\gamma(t) = \int_0^t q(s)q(s)ds$. As shown in Panel (b), the CLR-based resamples preserve the general structure and variability of the originals in Panel (a). In contrast, the SRVF-based resamples in Panel (c) include several invalid functions (highlighted in red), which are either non-monotonic or do not lie within the $[0, 1]$ interval.

This example illustrates a key limitation of the SRVF framework. The invalid samples occur because the Gaussian sampling imposes no constraint on the principal component scores. The same issue arises with Bayesian models using a GP prior on the tangent space of

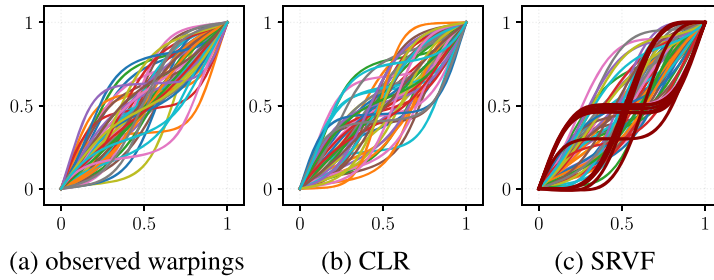


Fig. 1. Resampling using the CLR and SRVF models. (a) Observed random warplings. (b) Resampled warplings under the CLR model. (c) Resampled warplings under the SRVF model; some draws become invalid after the inverse SRVF map because the sampled SRD can violate the positivity constraint $q_\gamma(t) > 0$ at some t . Red bold curves indicate such invalid warplings.

the SRD. Although the tangent space faithfully captures the local geometry of the SRD manifold, it does not account for the manifold's global topology or boundary constraints. Because the GP prior is defined in this linear space, posterior proposals may map to functions on the SRD manifold that either wrap around, due to its spherical geometry, or cross the positivity boundary, potentially resulting in invalid warplings. In contrast, the CLR representation operates in an unconstrained linear inner-product space that preserves validity under sampling and inverse transformation.

2.2. New Bayesian framework on pairwise alignment

In this section, we first introduce the basic case in depth: a pairwise alignment of noisy signals, under our Bayesian framework.

2.2.1. Model specification and sampling algorithm

The main distinction from previous work lies in our decision to perform alignment directly in the original function space, rather than using the SRVF representation. [Srivastava and Klassen \(2016\)](#) demonstrated that representing functions via SRVF offers advantages for the registration problem: (1) it preserves isometry under warping and (2) mitigates the pinching effect during alignment. However, within our Bayesian framework, the prior distribution inherently penalizes undesirable behaviors such as pinching. As a result, we align the functions in their original space.

The likelihood model for pairwise alignment is given by:

$$y_2([t]) - (y_1 \circ \gamma)([t]) \mid \gamma \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I}), \quad (3)$$

where $y_1, y_2 \in \mathbb{L}^2([0, 1])$ represent noisy functional observations and are evaluated on a time grid of T points, i.e., $[t] = (0, \frac{1}{T-1}, \frac{2}{T-1}, \dots, 1) \in \mathbb{R}^T$, $\sigma^2 \in \mathbb{R}^+$, and $\mathbf{I} \in \mathbb{R}^{T \times T}$ is an identity matrix. Throughout the paper, we treat the noise variance σ^2 as a small fixed hyperparameter rather than updating it within the MCMC algorithm. For the simulation examples in [Section 2](#), the data-generating noise level is known, so σ^2 is chosen as a fixed fitting variance relative to that noise scale and reported in each simulation setting. A prior could be placed on σ^2 and updated within the same Bayesian framework, but this extension is not pursued here. We use this fixed- σ^2 strategy to emphasize the likelihood term, thereby allowing the data to drive the alignment by reducing the relative influence of prior regularization. We assume that the warping function γ and the observations y_1, y_2 are mutually independent a priori. In this setting, we impose a GP prior for the CLR-transformed warping function, $h_\gamma \in H(0, 1)$, as follows:

$$h_\gamma \sim \mathcal{GP}(\mu_h, \mathcal{K}_h),$$

where we denote $h_\gamma \triangleq \psi(\gamma)$ for notational convenience, and μ_h and $\mathcal{K}_h$ are the pre-determined mean function and covariance operator, respectively.

Remark 1 (Prevention of Pinching via Prior Regularization). By [Eq. \(2\)](#), $h = \psi(\gamma)$ and $\dot{\gamma}(t)$ is proportional to $e^{h(t)}$. An almost flat or almost vertical segment in γ requires $\dot{\gamma}(t)$ to be extremely small (i.e., close to 0) or extremely large on that interval. This can only happen when $h(t)$ has a very large magnitude on that interval. Under the GP prior for h , such values are in the tail of the prior. On the discretized time grid used in computation, the vector of values of h is multivariate normal, so vectors with very large magnitude have very small prior probability. Therefore, pCN proposals that create extreme local slopes in γ tend to have low acceptance unless the likelihood ratio increases substantially. With an additive-noise likelihood in the original space, the fit does not force exact pointwise matching, which further reduces the incentive for such extreme local deformation.

Conventionally, it can be defined using either the squared exponential (SE) or exponential (EXP) kernel, each characterized by two parameters: ν , $\ell > 0$, a scale factor and a length scale, respectively. Specifically, the covariance kernels are given by

$$\mathcal{K}_{\text{EXP}}(s, t) = \nu \exp\left(-\frac{|s - t|}{\ell}\right), \quad \text{and} \quad \mathcal{K}_{\text{SE}}(s, t) = \nu \exp\left(-\frac{(s - t)^2}{\ell}\right), \quad (4)$$

where $s, t \in [0, 1]$.

Since the warping function h_γ lies in an infinite-dimensional function space, both the prior and posterior must be defined as probability measures rather than conventional densities. We use the Z-mixture pCN algorithm to perform posterior sampling, following the earlier studies (Lu et al., 2017; Ma and Wu, 2025). This method enables efficient Metropolis-Hastings sampling under a GP prior by leveraging the likelihood, which can be interpreted as a finite-dimensional conditional density (Cotter et al., 2013). Furthermore, it allows consistent inference regardless of the resolution grid size, T , making it a robust method for function-space sampling. The posterior measure μ is assumed to be absolutely continuous with respect to the prior measure μ_0 , and the Radon-Nikodym derivative is given by:

$$\frac{d\mu}{d\mu_0}(h_\gamma) \propto \mathcal{L}(h_\gamma; y_1, y_2),$$

where $\mathcal{L}(h_\gamma; y_1, y_2)$ is the likelihood function. Given that $\gamma \triangleq \psi^{-1}(h_\gamma)$, the likelihood is specified as:

$$\mathcal{L}(h_\gamma; y_1, y_2) \propto \exp \left\{ -\frac{1}{2\sigma^2} \|y_2([t]) - (y_1 \circ \gamma)([t])\|^2 \right\}.$$

Here, $\|\cdot\|$ denotes the standard Euclidean norm on vectors, which is used throughout the paper. With a random weight $\beta \in (0, 1)$ drawn from a pre-given distribution, π_β , the proposal h'_γ by Z-mixture pCN is as follows:

$$h'_\gamma = \mu_h + \sqrt{1 - \beta^2} (h_\gamma - \mu_h) + \beta \xi, \quad \text{and} \quad \xi \sim \mathcal{GP}(0, \mathcal{K}_h).$$

This posterior proposal is accepted with a probability of $\min(1, \frac{\mathcal{L}(h'_\gamma; y_1, y_2)}{\mathcal{L}(h_\gamma; y_1, y_2)})$ which only requires likelihood functions. The choice of step-size parameters β plays a crucial role in the efficiency of the sampler. Smaller values of β (close to zero) lead to more conservative proposals, promoting stability and higher acceptance rates. Conversely, larger values (approaching one) yield more exploratory moves that facilitate faster mixing and may help the sampler escape from local modes in the posterior distribution.

In this study, we initialize h_γ at its prior mean μ_h . However, a good initialization can substantially accelerate convergence and reduce the risk of getting stuck in local optima. One possible choice for initialization is the F-R method (Kurtek et al., 2011), which, despite being designed for noise-free data, could still provide a useful indication of the general direction of alignment. The pseudo-code of this method is provided in Algorithm 7.

Algorithm 1: Pairwise alignment algorithm.

Input: Noisy pairwise data y_1, y_2 ; pre-given distribution π_β for β ; prior mean function and covariance operator for warplings, $\mu_h, \mathcal{K}_h$; error variance σ^2

Output: Posterior warping function samples $\{\gamma^{(m)}\}_{m=2}^{M+1}$

```

1  $h^{(1)} = \mu_h; \quad \gamma^{(1)} = \psi^{-1}(h^{(1)})$ 
2 for  $m = 1, 2, \dots, M$  do
3   Generate  $\beta_h \sim \pi_\beta$  and  $\xi \sim \mathcal{GP}(0, \mathcal{K}_h)$ 
4   Propose  $h' = \mu_h + \sqrt{1 - \beta_h^2} (h^{(m)} - \mu_h) + \beta_h \xi$ 
5   Update  $h^{(m+1)} = \begin{cases} h' & \text{with probability } \alpha_h = \min \left( 1, \frac{\mathcal{L}(h'; y_1, y_2)}{\mathcal{L}(h^{(m)}; y_1, y_2)} \right) \\ h^{(m)} & \text{with probability } 1 - \alpha_h \end{cases}$ 
6    $\gamma^{(m+1)} = \psi^{-1}(h^{(m+1)})$ 
7 end
```

2.2.2. Simulation illustration of pairwise alignment

We provide a numerical simulation to illustrate Algorithm 7 for registering a pair of noisy functions. The observed signals y_1 and y_2 are generated from two smooth functions f_1 and f_2 , respectively:

$$y_1 = f_1 + \epsilon_1, \quad \text{with} \quad f_1(t) = 0.4 e^{-\frac{(t-0.45)^2}{2 \cdot 0.05^2}} + 0.2 e^{-\frac{(t-0.72)^2}{2 \cdot 0.12^2}},$$

$$y_2 = f_2 + \epsilon_2, \quad \text{with} \quad f_2(t) = 0.3 e^{-\frac{(t-0.15)^2}{2 \cdot 0.03^2}} + 0.6 e^{-\frac{(t-0.5)^2}{2 \cdot 0.18^2}}, \quad \text{and}$$

$$\epsilon_1([t]), \epsilon_2([t]) \sim \mathcal{N}(\mathbf{0}, 0.08^2 \mathbf{I}),$$

where the time grid $[t] = (0, 1, 2, \dots, 100)/100$. We fix the noise variance at $\sigma^2 = 0.05^2$ and use a squared exponential kernel for the GP prior on the CLR-transformed warping $h_\gamma = \psi(\gamma)$, with $\mathcal{K}_h(s, t) = 100 \exp\{-(s-t)^2/10\}$. Kernel hyperparameters are fixed using prior predictive checks under the full estimation model in Section 2.4.2. Details are deferred to that section. Fig. 2(a) shows the latent signals f_1, f_2 and the noisy observations y_1, y_2 .

Fig. 2(b) shows the aligned signals using the F-R method and our proposed Bayesian method, respectively. Each column shows the aligned signals and the corresponding warping. Especially, for our method, the estimation of the warping function is defined by posterior samples, i.e., $\psi^{-1} \left(\frac{1}{M-m_b+1} \sum_{m=m_b}^M h_\gamma^{(m)} \right)$, where $m_b \geq 2$ denotes the starting index after burn-in, used to compute the posterior

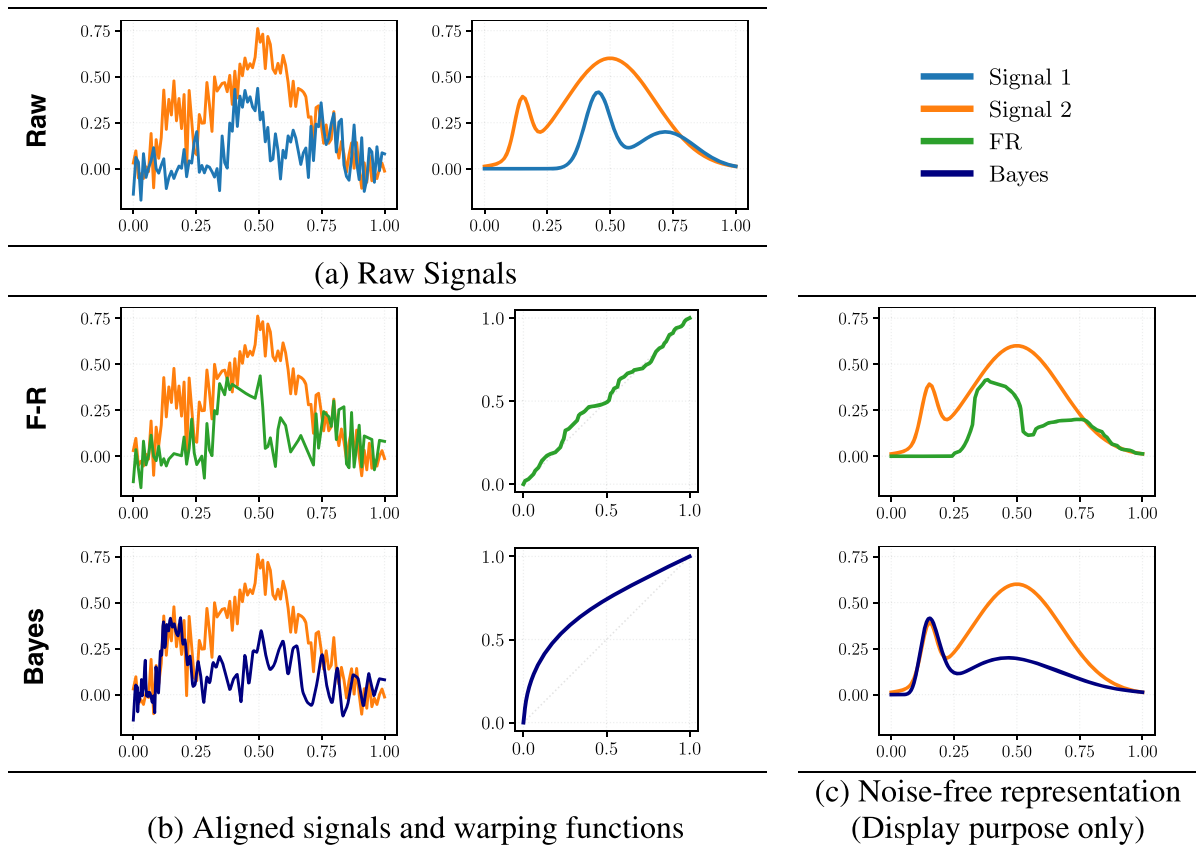


Fig. 2. Raw signals and alignment result comparison. (a) Raw signals with noise and noise-free representations. (b) Alignment results by F-R (upper, green) and our proposed method (lower, dark blue). (c) Corresponding aligned noise-free signals, shown for visual comparison only (not used in estimation). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

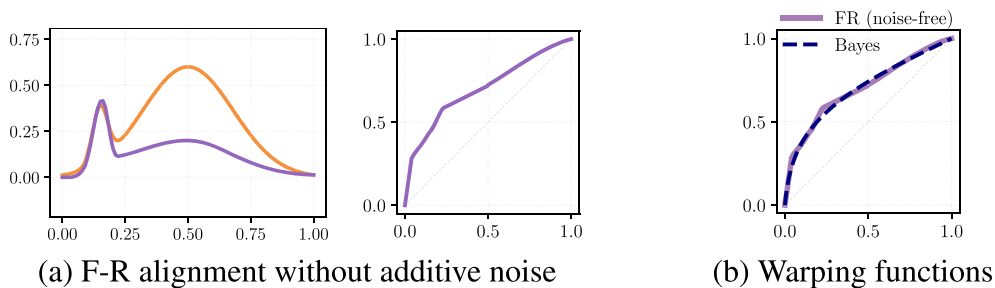


Fig. 3. F-R alignment noise-free signals and comparison. (a) Aligned f_1 (purple) and f_2 (orange) and the estimated warping function via F-R. (b) Comparison of warping functions to our proposed method (dark blue). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

average from samples m_b to M . As shown in the second column of each row in the figure, the F-R method results in poor alignment of y_1 to y_2 , as it barely deforms the original signal and produces a less smooth warping function. In contrast, our Bayesian method yields a smoother warping function and achieves visibly desirable alignment with y_2 . This estimation relies on stable MCMC sampling, characterized by an acceptance rate of 9.1% for h_γ and an Effective Sample Size (ESS) of 380.7 per minute.

For a more transparent comparison of the alignment between the F-R and our method, Fig. 2(c) presents the noise-free representation of aligned signals. The upper and lower panels correspond to the alignments via F-R and our method, respectively. As shown, our method aligns each peak of f_1 and f_2 accordingly, while F-R does not. To provide a sanity check of our alignment result, we provide the alignment via the F-R but on noise-free signals, f_1 and f_2 . The alignment results by F-R with f_1 and f_2 , as well as the warping function, are shown in Fig. 3.

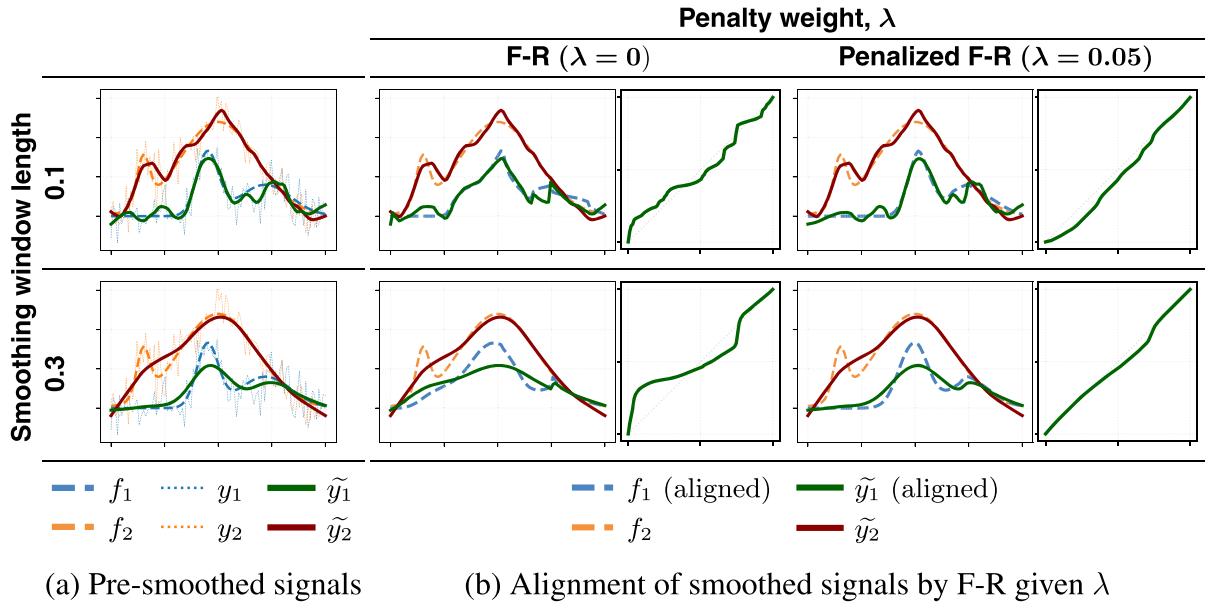


Fig. 4. Alignment results of the F-R method with pre-smoothing and regularization. (a) Pre-smoothed raw signals arranged by row, using smoothing window lengths of 0.1 and 0.3, respectively. Here, $\tilde{y}_1$ and $\tilde{y}_2$ denote smoothed y_1 and y_2 , respectively. (b) Alignment results across a grid: rows vary by smoothing level (0.1, 0.3), and columns vary by penalty weight (0, 0.05).

This simple illustration effectively demonstrates the absence of the *pinching effect*, which is naturally curbed by the prior on the warping function. Moreover, it highlights the method's robustness to additive noise, achieving near-perfect alignment between the two noisy signals. This robustness arises from avoiding the use of SRVFs of the observed functions, where additive noise causes disproportionately large distortions. In Section 3.1.2, we will present an extreme case in which small additive noise in the original function space overwhelms the main signal in the SRVF space.

We also acknowledge that, in the F-R method, robustness to additive noise can be improved through two approaches: (1) pre-smoothing the signals and (2) applying regularization. To illustrate this, we present the alignments obtained under varying levels of these two factors, as shown in Fig. 4. Each row in Panel (a) displays the smoothed observed functions (solid curves) produced by LOWESS (locally weighted scatterplot smoothing (Cleveland and Devlin, 1988)) with smoothing window lengths of 0.1 and 0.3, respectively. In the domain $[0, 1]$, this means that smoothing at each point in the signal is performed using a neighborhood of length 0.1 or 0.3, respectively. Panel (b) shows, in columns, the alignments of the corresponding pre-smoothed signals from Panel (a) obtained via penalized F-R alignment (Srivastava and Klassen, 2016) with penalty weights of $\lambda = 0$ and 0.05, respectively.

When the smoothing window length is 0.1 (upper row), the signals are under-smoothed, resulting in numerous spurious features arising from the additive noise. With $\lambda = 0$, the F-R alignment matches the first peak of f_1 to the second peak of f_2 . In contrast, the F-R method under noise-free settings, shown in Fig. 3(a), matches each peak in f_1 and f_2 as our method does; their warping functions are compared in Panel (b), and they are comparably identical, but ours is smoother. As in the second column of Fig. 4(b), regularization with $\lambda = 0.05$ does not improve the situation, as it only makes the estimated warping closer to the identity warping, resulting in a mismatch of the peaks in f_1 and f_2 . With a smoothing window length of 0.3 (bottom row), the signals are over-smoothed, and regardless of the penalty, this also mismatches the peaks. This suggests that the robustness gained by traditional methods may not be sufficient, implying that a more systematic approach is needed.

2.3. New Bayesian framework on multiple alignment

The pairwise alignment algorithm can be extended to the multiple alignment setting. The key difference from the pairwise case is the introduction of an unknown template function, g , to which all signals are aligned.

2.3.1. Model specification and sampling algorithm

Given the N independent observed noisy signals, $\{y_i\}_{i=1}^N$, we align each signal y_i to the shared, unknown template function, g , using its own warping function, γ_i , for $i = 1, \dots, N$. Although we conceptually align each observed signal toward g , the model is defined with the inverse of the warping function, γ_i^{-1} , which is applied to g . Specifically, we seek a smooth function g such that $g \circ \gamma_i^{-1}$ closely approximates y_i for each i . This leads to the following likelihood formulation:

$$y_i([t]) - (g \circ \gamma_i^{-1})([t]) \mid g, \gamma_i \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I}) \quad \text{for } i = 1, \dots, N, \quad (5)$$

where $\mathbf{0} \in \mathbb{R}^T$ is a zero vector and $\mathbf{I} \in \mathbb{R}^{T \times T}$ is an identity matrix.

Note that Equation (5) represents a simplified model without scale and translation. To resolve the warping ambiguity conditional on these fixed amplitude parameters, we impose the constraint that the CLR-transformed warping function satisfies $\mathbb{E}[h_{\gamma_i}(t)] = 0$ for all $t \in [0, 1]$. This ensures that the set $\{h_{\gamma_i}\}$ is centered at $h_{\gamma_{id}}(t) = 0$, where $\gamma_{id}(t) = t$ is the identity warping. Thus, this condition is used to fix the common warping ambiguity by centering the warpings at the identity. It does not imply that g is uniquely identifiable without this centering convention.

The CLR-transformed warping functions $\{h_{\gamma_i}\}_{i=1}^N$ are modeled as independent and identically distributed Gaussian processes with mean function μ_h and covariance operator $\mathcal{K}_h$. The template function g is also given a Gaussian process prior with mean μ_g and covariance operator $\mathcal{K}_g$:

$$\{h_{\gamma_i}\}_{i=1}^N \sim \mathcal{GP}(\mu_h, \mathcal{K}_h) \quad \text{for } i = 1, \dots, N, \quad \text{and} \quad g \sim \mathcal{GP}(\mu_g, \mathcal{K}_g).$$

Let μ_0 and μ denote the prior and posterior distributions, respectively. These are probability measures over the product space $\mathbb{L}^2([0, 1])^{N+1}$ corresponding to the collection $(h_{\gamma_1}, \dots, h_{\gamma_N}, g)$. The posterior measure μ is assumed to be absolutely continuous with respect to the prior μ_0 and is characterized via the Radon-Nikodym derivative.

$$\frac{d\mu}{d\mu_0}(h_{\gamma_1}, \dots, h_{\gamma_N}, g) \propto \mathcal{L}(h_{\gamma_1}, \dots, h_{\gamma_N}, g; y_1, \dots, y_N),$$

where $\mathcal{L}(h_{\gamma_1}, \dots, h_{\gamma_N}, g; y_1, \dots, y_N)$ is the likelihood. Leaving $\gamma_i \triangleq \psi^{-1}(h_{\gamma_i})$, the likelihood is given as follows:

$$\mathcal{L}(h_{\gamma_1}, \dots, h_{\gamma_N}, g; y_1, \dots, y_N) = \prod_{i=1}^N \mathcal{L}(h_{\gamma_i}, g; y_i),$$

where $\mathcal{L}(h_{\gamma_i}, g; y_i) \propto \exp \left\{ -\frac{1}{2\sigma^2} \left\| y_i([t]) - (g \circ \gamma_i^{-1})([t]) \right\|^2 \right\}$.

As in the pairwise alignment case, we obtain posterior samples of $\{h_{\gamma_i}\}$ and g using the Z-mixture pCN scheme within a Metropolis-Hastings step embedded in a Gibbs sampler, as described in Algorithm 16. In addition to these updates, each MCMC iteration includes a centering step to implement the warping-centering convention described below.

Algorithm 2: Multiple alignment algorithm.

Input: Data $\{y_i\}_{i=1}^N$; pre-given distribution $\pi_{\beta_h}, \pi_{\beta_g}$ for $\beta_h, \beta_g \in [0, 1]$; prior mean and covariance operator for $h_{1:N}$: $\mu_h, \mathcal{K}_h$; prior mean and covariance operator for g : $\mu_g, \mathcal{K}_g$; noise variance σ^2

Output: Estimated functions $\{\gamma_{1:N}^{(m)}\}_{m=2}^{M+1}$

```

1   $g^{(1)} = \mu_g$ 
2   $h_i^{(1)} = \mu_h$ ;  $\gamma_i^{(1)} = \psi^{-1}(h_i^{(1)})$  for  $i = 1, \dots, N$ 
3  for  $m = 1, 2, \dots, M$  do
4      for  $i = 1, 2, \dots, N$  do
5          Generate  $\beta_h \sim \pi_{\beta_h}$  and  $\xi \sim GP(0, \mathcal{K}_h)$ 
6          Propose  $h'_i = \mu_h + \sqrt{1 - \beta_h^2} (h_i^{(m)} - \mu_h) + \beta_h \xi$ 
7          Update  $h_i^{(m+1)} = \begin{cases} h'_i & \text{with probability } \alpha_h = \min \left( 1, \frac{\mathcal{L}(h'_i, g^{(m)}; y_i)}{\mathcal{L}(h_i^{(m)}, g^{(m)}; y_i)} \right) \\ h_i^{(m)} & \text{with probability } 1 - \alpha_h \end{cases}$ 
8           $\gamma_i^{(m+1)} = \psi^{-1}(h_i^{(m+1)})$ 
9      end
10      $\bar{\gamma} = \psi^{-1} \left( \frac{1}{N} \sum_{i=1}^N h_i^{(m+1)} \right)$ 
11      $\gamma_{1:N}^{(m+1)} \leftarrow \gamma_{1:N}^{(m+1)} \circ \bar{\gamma}^{-1}$ ;  $g^{(m)} \leftarrow g^{(m)} \circ \bar{\gamma}^{-1}$ 
12      $h_{1:N}^{(m+1)} \leftarrow \psi(\gamma_{1:N}^{(m+1)})$ 
13     Generate  $\beta_g \sim \pi_{\beta_g}$  and  $\zeta \sim GP(0, \mathcal{K}_g)$ 
14     Propose  $g' = \mu_g + \sqrt{1 - \beta_g^2} (g^{(m)} - \mu_g) + \beta_g \zeta$ 
15     Update  $g^{(m+1)} = \begin{cases} g' & \text{with probability } \alpha_g = \min \left( 1, \prod_{i=1}^N \frac{\mathcal{L}(h_i^{(m+1)}, g'; y_i)}{\mathcal{L}(h_i^{(m+1)}, g^{(m)}; y_i)} \right) \\ g^{(m)} & \text{with probability } 1 - \alpha_g \end{cases}$ 
16 end
```

Remark 2. ([Warping ambiguity and centering]) The model likelihood in Eq. (5) remains invariant under a simultaneous composition by any common warping $\bar{\gamma}^{-1}$, since

$$(g \circ \bar{\gamma}^{-1}) \circ (\gamma_i \circ \bar{\gamma}^{-1})^{-1} = g \circ \gamma_i^{-1}.$$

This invariance introduces a global warping ambiguity, so the pair $(g, \{\gamma_i\})$ is identifiable only up to a common warping transformation. The condition $\mathbb{E}[\psi(\gamma_i)] = 0$, equivalently centering the warpings at $\psi(\gamma_{id})(t) = 0$, is used as a centering convention to select a centered representative within each equivalence class. It does not imply that g and $\{\gamma_i\}$ are uniquely identifiable without this convention. For

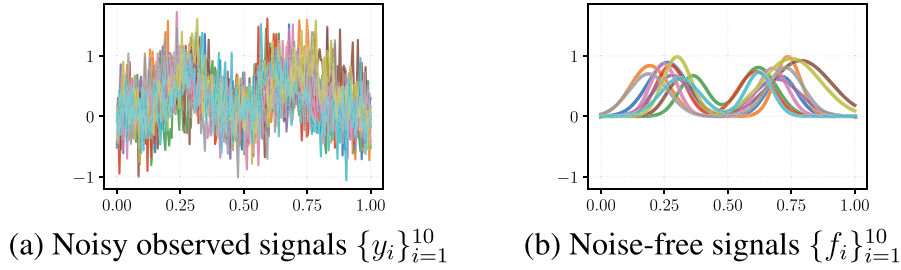


Fig. 5. Ten simulated signals for multiple function alignment.

the full model in Eq. (1), the additional constraints $\mathbb{E}[c_i] = 1$ and $\mathbb{E}[e_i] = 0$ are used to fix the global scale and translation components.

The centering convention is implemented using the mean-preserving property of the CLR representation. Specifically, for a random warping γ_i ,

$$\mathbb{E}[\psi(\gamma_i \circ \gamma)] = \psi(\bar{\gamma}_{\text{clr}} \circ \gamma),$$

where $\psi(\bar{\gamma}_{\text{clr}}) = \mathbb{E}[\psi(\gamma_i)]$ is the CLR-defined population mean with the operator ψ given in Eq. (2) (Ma and Wu, 2025). This property allows us to shift the center to the identity by imposing the constraint $\mathbb{E}[\psi(\gamma_i)] = 0$. In practice, this population mean is approximated by the empirical average

$$\bar{\gamma} = \psi^{-1} \left(\frac{1}{N} \sum_{i=1}^N \psi(\gamma_i) \right).$$

At each MCMC iteration, we re-center the warping functions and the template function by composing with the inverse of the empirical mean:

$$\gamma_i \leftarrow \gamma_i \circ \bar{\gamma}^{-1}, \quad g \leftarrow g \circ \bar{\gamma}^{-1}.$$

This operation shifts the center of the current set of warping functions to the identity. The resulting posterior samples are therefore interpreted as centered representatives within their equivalence classes.

2.3.2. Simulation illustration of multiple alignment

To demonstrate the performance of Algorithm 16 for multiple alignment, we present a simulation study using synthetic functional data. To generate the simulated data $\{y_i\}_{i=1}^{10}$, each latent signal $f_i(t)$ is constructed as a sum of two Gaussian peaks:

$$f_i(t) = \sum_{j=1}^2 a_{ij} \cdot \exp \left\{ -\frac{(t - \mu_{ij})^2}{2\sigma_{ij}^2} \right\},$$

and the observed function is given by $y_i = f_i + \epsilon_i$, where $\epsilon_i \sim \mathcal{N}(\mathbf{0}, 0.35^2 \mathbf{I})$. For each i , the amplitudes a_{i1}, a_{i2} are sampled independently from $\text{Unif}([0.6, 1])$; the centers μ_{i1}, μ_{i2} from $\text{Unif}([0.18, 0.4])$ and $\text{Unif}([0.58, 0.8])$, respectively; and the standard deviations σ_{i1}, σ_{i2} from $\text{Unif}([0.05, 0.08])$ and $\text{Unif}([0.05, 0.12])$, respectively.

Fig. 5(a) and (b) show the ten simulated signals with additive noise (i.e., $\{y_i\}_{i=1}^{10}$) and without additive noise (i.e., $\{f_i\}_{i=1}^{10}$), respectively.

Given these signals, the alignment results of the comparable models are presented in Fig. 6. Each row shows the results obtained using one of four approaches: (1) the F-R method, (2) the SRVF-based Bayesian method using the SRD warping representation, (3) the SRVF-based Bayesian method with the CLR warping representation, and (4) our proposed method. Panel (a) displays the aligned signals and the corresponding warping functions for each method in rows. As shown in the first row, the F-R approach shows limited success in aligning the noisy signals y_i 's, with its estimated warping functions remaining close to the identity transformation. The SRVF-based Bayesian methods (in the second and third rows) appear to align the signals; however, a closer examination at the individual level reveals that the alignment is not adequate. In contrast, our proposed method, shown in the final row, clearly outperforms the other methods even upon visual inspection.

Panel (b) showcases the noise-free representation of aligned signals by each method (in rows). It is presented only for comparison purposes, and f_i 's are not in the alignment process at all. As shown, it becomes more evident that our method aligns the signals better than the others.

For the prior selections, our proposed Bayesian model employs $g \sim \mathcal{GP}(0, \mathcal{K}_g)$ where $\mathcal{K}_g$ is a squared exponential kernel for with scale factor $\nu = 1$ and length scale $\ell = 0.01$ as described in Eq. (4). This choice of ℓ encourages smooth estimation of the template g . A larger ℓ in $\mathcal{K}_g$ enforces smoother estimates, offering a tuning mechanism for controlling smoothness in g . The time-warping prior is $h_i \sim (0, \mathcal{K}_h)$ where $\mathcal{K}_h$ is an exponential kernel with $\nu = 0.5$ and $\ell = 5$. At last, the prior of the noise, we set $\epsilon_i \sim \mathcal{N}(0, 0.03^2)$.

The convergence of the MCMC algorithms in Bayesian methods (2), (3) and (4) can be checked by simply displaying the log-likelihood, $\log \mathcal{L}(h_{\gamma_1}^{(m)}, \dots, h_{\gamma_N}^{(m)}, g^{(m)}; y_1, \dots, y_N)$ versus the MCMC iteration, m for each case shown in Fig. 7. From this, our method

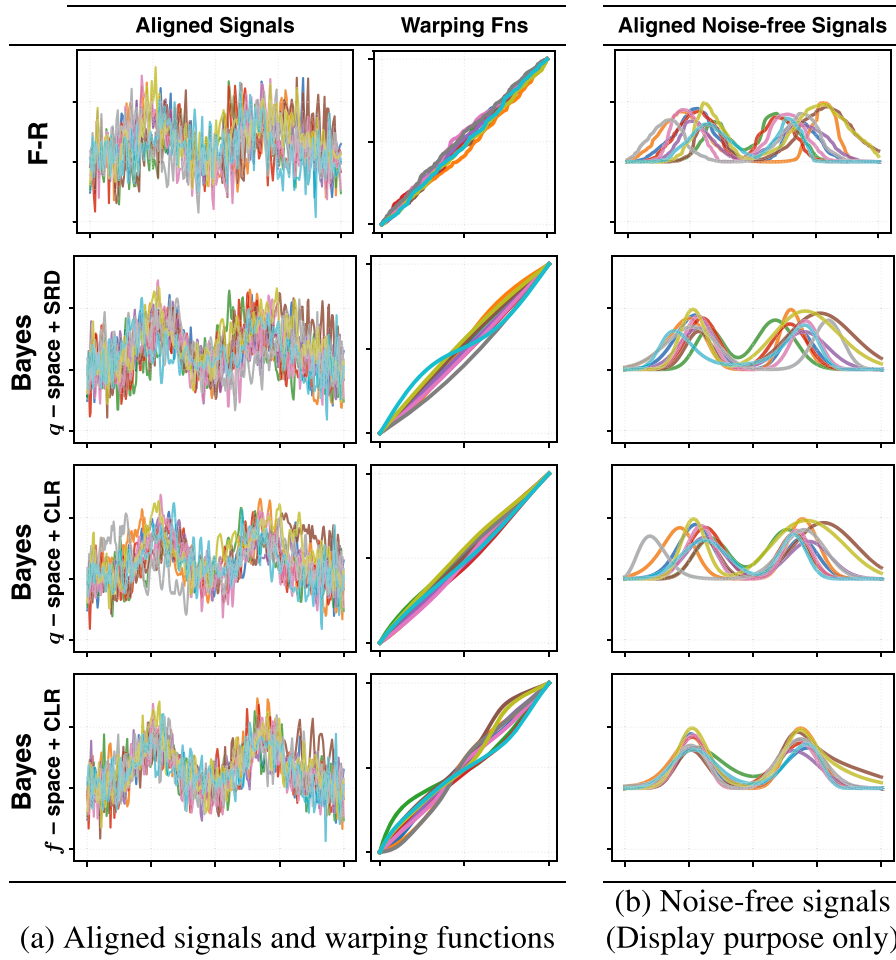


Fig. 6. Alignment Comparison. (a) The alignment results, in rows, for (1) F-R, (2) Bayes (q -space + SRD), (3) Bayes (q -space + CLR), and (4) our proposed model. (b) Corresponding aligned noise-free signals, shown for visual comparison only (not used in estimation).

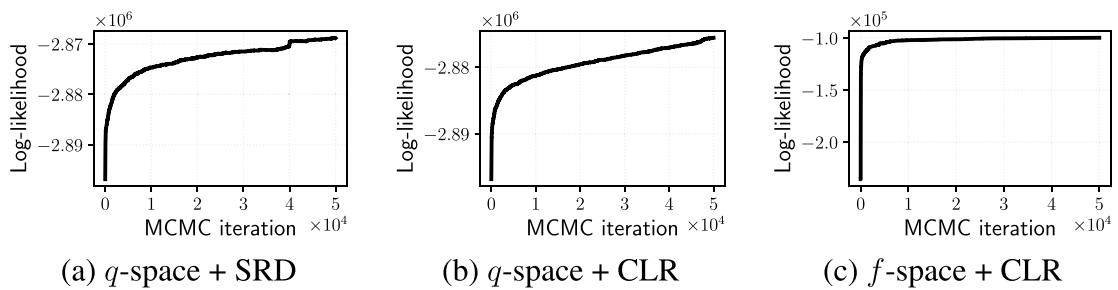


Fig. 7. Log-likelihood vs MCMC iteration for Bayesian models.

demonstrates greater efficiency in handling noisy signals. Panels (a) and (b), SRVF-based Bayesian methods, the SRVF inherently amplifies additive noise since it is derived from γ exhibit persistent increases in log-likelihood without apparent convergence even after 50,000 iterations. In comparison, our method in Panel (c) achieves convergence to the posterior distribution in fewer than 10,000 iterations. This rapid convergence is supported by the following diagnostic metrics; the proposed algorithm maintained an average acceptance rate of approximately 22.8% for h_i 's and 23.8% for g , with ESS values exceeding 110 per minute, confirming efficient mixing relative to the competing methods.

For reference, Fig. 8 shows the F-R alignment applied to noise-free signals, providing an approximate benchmark of performance without noise. We observe that the alignment achieved by our method remains comparable to this benchmark.

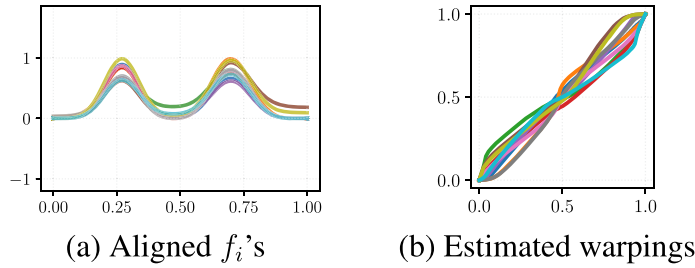


Fig. 8. Aligned noise-free signals and estimated warpings via F-R method without additive noise.

Table 2

Performance metrics for different alignment methods.

	Least Squares	Pairwise Corr	Sobolev Least Squares
F-R	1.45	0.81	3.05
q -space + SRD	0.88	1.19	0.92
q -space + CLR	1.44	1.17	1.52
f -space + CLR	0.61	1.92	0.65
F-R (noise-free)	0.51	2.02	0.21

We also evaluate the alignment performances of the methods using three criteria, adopted from (Srivastava et al., 2011). To define each criterion, we denote the original and aligned signals by f_i and $\tilde{f}_i$, respectively, for $i = 1, \dots, N$. To make the content self-contained, we briefly describe them as follows:

- The first criterion, *least squares*, measures the synchronization of the aligned signals using the $\mathbb{L}^2$ distance as follows:

$$\frac{1}{N} \sum_{i=1}^N \frac{\int (\tilde{f}_i(t) - \frac{1}{N-1} \sum_{j \neq i} \tilde{f}_j(t))^2 dt}{\int (f_i(t) - \frac{1}{N-1} \sum_{j \neq i} f_j(t))^2 dt}.$$

This quantity is the mean of the cross-sectional variance of the aligned functions relative to the original value. Smaller values indicate better alignment.

- The second criterion, *pairwise correlation*, measures the pairwise Pearson correlation between signals:

$$\frac{\sum_{i \neq j} cc(\tilde{f}_i, \tilde{f}_j)}{\sum_{i \neq j} cc(f_i, f_j)},$$

where $cc(\cdot, \cdot)$ denotes the Pearson correlation between two signals. In this case, larger values indicate better alignment.

- The third criterion, *Sobolev least squares*, uses the Sobolev distance, which measures the distance between the first derivatives of the signals:

$$\frac{\sum_{i=1}^N \int (\dot{\tilde{f}}_i(t) - \frac{1}{N} \sum_{j=1}^N \dot{\tilde{f}}_j(t))^2 dt}{\sum_{i=1}^N \int (\dot{f}_i(t) - \frac{1}{N} \sum_{j=1}^N \dot{f}_j(t))^2 dt}.$$

This quantity measures the ratio of the total cross-sectional variances of the first derivatives of the aligned and original signals. As with the first criterion, a smaller value indicates better alignment.

Table 2 lists these criteria in columns for different methods in rows, and we assure our proposed method (fourth row) outperforms the others. Compared to the noise-free alignment using the F-R metric (last row), our method achieves comparable performance.

2.4. Signal estimation

While the template function g plays a central role in aligning multiple signals, it may not correspond to an actual underlying signal in real-world data. For example, different signals may exhibit varying numbers of peaks or differences in peak height and ordering, rendering the concept of a single, shared underlying signal unrealistic. Nonetheless, when the observations do follow the generative model described in Eq. (1), it becomes meaningful to investigate whether g can be accurately estimated from the noisy, warped observations. This section focuses on evaluating the ability of our Bayesian framework to recover such a template under the assumed model, serving as a step toward understanding its consistency properties.

2.4.1. Generative model and consistency algorithm

In this section, we investigate the empirical behavior of our Bayesian estimation procedure, focusing on its ability to recover the latent template and other components under increasingly rich observational settings. This study serves as a practical consistency check, helping to assess how well the posterior concentrates around the ground truth.

Given that the data generation procedure follows Eq. (1), the statistical model of discretized observations is as follows:

$$y_i([t]) = c_i g \circ \gamma_i^{-1}([t]) + e_i + \epsilon_i([t]), \quad \text{and} \quad \epsilon_i([t]) \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I}) \quad \text{for} \quad i = 1, \dots, N,$$

where $\mathbf{0} \in \mathbb{R}^T$ is a zero vector and $\mathbf{I} \in \mathbb{R}^{T \times T}$ is an identity matrix. Then we place the priors as follows:

$$\begin{aligned} h_1, \dots, h_N &\stackrel{\text{iid}}{\sim} \mathcal{GP}(0, \mathcal{K}_h), & g &\sim \mathcal{GP}(0, \mathcal{K}_g), \\ e_1, \dots, e_N &\stackrel{\text{iid}}{\sim} \mathcal{N}(0, \sigma_e^2), & \text{and} & \\ c_1, \dots, c_N &\stackrel{\text{iid}}{\sim} \text{Gamma}(\text{shape: } \alpha_c, \text{scale: } \theta_c). \end{aligned}$$

To address model identifiability, we impose centering constraints on the scale, translation, and warping components. We set $\mathbb{E}[c_i] = 1$ and $\mathbb{E}[e_i] = 0$ to fix the global scale and shift. For the phase component, following the CLR-centering approach of Ma and Wu (2025), we require $\mathbb{E}[h_{\gamma_i}(t)] = 0$ for all $t \in [0, 1]$ to center the warping functions in $H(0, 1)$.

Let μ_0 and μ denote the prior and posterior distributions, respectively. These are probability measures on the product space $\mathbb{L}^2([0, 1])^{N+1} \times \mathbb{R}^{2N}$, corresponding to the collection $(h_{\gamma_{1:N}}, g, c_{1:N}, e_{1:N})$. The posterior μ is assumed to be absolutely continuous with respect to the prior μ_0 , and the Radon-Nikodym derivative is given by

$$\frac{d\mu}{d\mu_0}(h_{\gamma_{1:N}}, g, c_{1:N}, e_{1:N}) \propto \mathcal{L}(h_{\gamma_{1:N}}, g, c_{1:N}, e_{1:N}; y_1, \dots, y_N),$$

where the likelihood with $\gamma_i \triangleq \psi^{-1}(h_{\gamma_i})$ is as follows:

$$\mathcal{L}(h_{\gamma_{1:N}}, g, c_{1:N}, e_{1:N}; y_1, \dots, y_N) = \prod_{i=1}^N \mathcal{L}(h_{\gamma_i}, g, c_i, e_i; y_i),$$

$$\text{and } \mathcal{L}(h_{\gamma_i}, g, c_i, e_i; y_i) \propto \exp \left\{ -\frac{1}{2\sigma^2} \left\| y_i([t]) - (e_i + c_i(g \circ \gamma_i^{-1})([t])) \right\|^2 \right\}.$$

The MCMC scheme in Algorithm 19 largely follows that of the previous multiple-function alignment setup, yet in this extended model, we additionally update the scale terms c_i and the translation terms e_i . We begin with the scale update. Since there is no conjugate prior for c_i , we sample each c_i using a Metropolis-Hastings step with a Gamma prior. To ensure positivity and enable unconstrained proposals, we perform updates in log-space:

$$c'_i \sim \text{LogN}(u_i, \sigma_{\text{prop}}^2), \quad \text{where } u_i = \log c_i.$$

Since the proposal is made in log-space with a symmetric normal distribution, the proposal ratio cancels in the Metropolis-Hastings step. We transform the Gamma prior on $c_i \sim \text{Gamma}(\alpha_c, \theta_c)$ using the change-of-variables formula:

$$\pi_U(u_i) = \pi_C(e^{u_i}) \cdot e^{u_i} = \frac{1}{\Gamma(\alpha_c) \theta_c^{\alpha_c}} \cdot e^{\alpha_c u_i} \cdot \exp \left(-\frac{e^{u_i}}{\theta_c} \right),$$

where π_C is the Gamma density. Then, the acceptance probability for the proposal, i.e., $u' = \log c'_i$, is given by

$$\alpha_c = \min \left(1, \frac{\mathcal{L}(c'_i, h_{\gamma_i}, e_i, g; y_i) \pi_U(u'_i)}{\mathcal{L}(c_i, h_{\gamma_i}, e_i, g; y_i) \pi_U(u_i)} \right).$$

We now turn to the update of the translation terms e_i , which admit closed-form updates due to Gaussian conjugacy. At a fixed MCMC iteration, the posterior update for e_i follows the normal-normal conjugate form:

$$e_i \sim \mathcal{N}(\mu_{e|y}, \sigma_{e|y}^2),$$

where

$$\mu_{e|y} = \sigma_{e|y}^2 \left(\frac{1}{\sigma_e^2} \cdot \mathbf{1}^\top (y_i([t]) - c_i(g \circ \gamma_i^{-1})([t])) \right), \quad \text{and} \quad \sigma_{e|y}^2 = \left(\frac{1}{\sigma_e^2} + \frac{T}{\sigma^2} \right)^{-1}.$$

2.4.2. Simulation illustration

We illustrate the proposed Bayesian framework using a synthetic dataset generated under the linear model described in Section 2.4.1 to assess whether our method can accurately recover the underlying template function g . The data is generated according to the model $y_i = e_i + c_i(g \circ \gamma_i^{-1}) + \epsilon_i$, where the template function is defined as

$$g(t) = \sum_{j=1}^3 a_j \exp \left(-\frac{(t - \mu_j)^2}{2\sigma_j^2} \right),$$

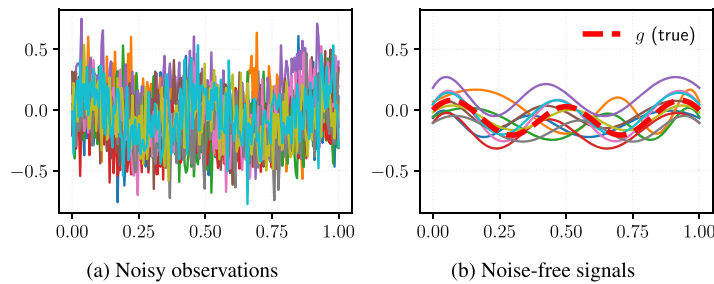
with parameters $(a_1, a_2, a_3) = (1.5, 1.2, 1.5)$, $(\mu_1, \mu_2, \mu_3) = (0.06, 0.5, 0.94)$, and $(\sigma_1, \sigma_2, \sigma_3) = (0.2, 0.15, 0.2)$. The additive noise is drawn as $\epsilon_i([t]) \stackrel{\text{iid}}{\sim} \mathcal{N}(\mathbf{0}, 0.2^2 \mathbf{I})$, while the translation and scale parameters follow $e_i \stackrel{\text{iid}}{\sim} \mathcal{N}(0, 0.1^2)$ and $c_i \stackrel{\text{iid}}{\sim} \text{Gamma}(10, 0.1)$, respectively. In Fig. 9, Panel (a) shows the noisy observations $\{y_i\}_{i=1}^{10}$ and Panel (b) displays noise-free representation of y_i 's along with underlying signal g (dotted red).

Algorithm 3: Signal estimation.

Input: Data $\{y_i\}_{i=1}^N$; pre-given distribution $\pi_{\beta_h}, \pi_{\beta_g}$; prior mean and covariance for h : $\mu_h, \mathcal{K}_h$; prior mean and covariance for g : $\mu_g, \mathcal{K}_g$; noise variance σ^2 ; prior variance for translation e_i : σ_e^2 ; transformed prior for scale π_U (induced by $c_i \sim \text{Gamma}(\alpha_c, \theta_c)$)

Output: Estimated Underlying Signal $\{g^{(m)}\}_{m=2}^{M+1}$

- 1 $g^{(1)} = \mu_g$; $h_i^{(1)} = \mu_h$; $\gamma_i^{(1)} = \psi^{-1}(h_i^{(1)})$; $c_i^{(1)} = 1$; $e_i^{(1)} = 0$ for $i = 1, \dots, N$
- 2 **for** $m = 1, 2, \dots, M$ **do**
- 3 **for** $i = 1, 2, \dots, N$ **do**
- 4 Generate $\beta_h \sim \pi_{\beta_h}$ and $\xi \sim GP(0, \mathcal{K}_h)$
- 5 Propose $h'_i = \mu_h + \sqrt{1 - \beta_h^2} (h_i^{(m)} - \mu_h) + \beta_h \xi$
- 6 Update $h_i^{(m+1)} = \begin{cases} h'_i & \text{with probability } \alpha_h = \min\left(1, \frac{\mathcal{L}(h'_i, e_i^{(m)}, c_i^{(m)}, g^{(m)}; y_i)}{\mathcal{L}(h_i^{(m)}, e_i^{(m)}, c_i^{(m)}, g^{(m)}; y_i)}\right) \\ h_i^{(m)} & \text{with probability } 1 - \alpha_h \end{cases}$
- 7 $\gamma_i^{(m+1)} = \psi^{-1}(h_i^{(m+1)})$
- 8 Calculate $\sigma_{e|y}^2 = \left(\frac{1}{\sigma_e^2} + \frac{T}{\sigma^2}\right)^{-1}$ and $\mu_{e|y} = \sigma_{e|y}^2 \frac{1^\top}{\sigma^2} \left(y_i - c_i^{(m)} (g^{(m)} \circ \gamma_i^{(m+1)})^{-1}([t])\right)$
- 9 Update $e_i^{(m+1)} \sim \mathcal{N}(\mu_{e|y}, \sigma_{e|y}^2)$
- 10 Propose $u'_i \sim \mathcal{N}(u_i^{(m)}, \sigma_{\text{prop}}^2)$ where $u_i^{(m)} = \log c_i^{(m)}$
- 11 Set $c'_i = \exp(u'_i)$
- 12 Update $c_i^{(m+1)} = \begin{cases} c'_i & \text{w.p. } \alpha_c = \min\left(1, \frac{\mathcal{L}(h_i^{(m+1)}, e_i^{(m+1)}, c'_i, g^{(m)}; y_i)}{\mathcal{L}(h_i^{(m+1)}, e_i^{(m+1)}, c_i^{(m)}, g^{(m)}; y_i)} \frac{\pi_U(u'_i)}{\pi_U(u_i^{(m)})}\right) \\ c_i^{(m)} & \text{w.p. } 1 - \alpha_c \end{cases}$
- 13 **end**
- 14 $\bar{c} = \sum_{i=1}^N c_i^{(m+1)} / N$; $\bar{\gamma} = \psi^{-1}\left(\frac{1}{N} \sum_{i=1}^N h_i^{(m+1)}\right)$; $c_{1:N}^{(m+1)} \leftarrow c_{1:N}^{(m+1)} / \bar{c}$
- 15 $\gamma_{1:N}^{(m+1)} \leftarrow \gamma_{1:N}^{(m+1)} \circ \bar{\gamma}^{-1}$; $g^{(m)} \leftarrow \bar{c} g^{(m)} \circ \bar{\gamma}^{-1}$; $h_{1:N}^{(m+1)} \leftarrow \psi(\gamma_{1:N}^{(m+1)})$
- 16 Generate $\beta_g \sim \pi_{\beta_g}$ and $\zeta \sim GP(0, \mathcal{K}_g)$
- 17 Propose $g' = \mu_g + \sqrt{1 - \beta_g^2} (g^{(m)} - \mu_g) + \beta_g \zeta$
- 18 Update $g^{(m+1)} = \begin{cases} g' & \text{with probability } \alpha_g = \min\left(1, \prod_{i=1}^N \frac{\mathcal{L}(h_i^{(m+1)}, e_i^{(m+1)}, c_i^{(m+1)}, g'; y_i)}{\mathcal{L}(h_i^{(m+1)}, e_i^{(m+1)}, c_i^{(m+1)}, g^{(m)}; y_i)}\right) \\ g^{(m)} & \text{with probability } 1 - \alpha_g \end{cases}$
- 19 **end**

**Fig. 9.** Ten simulated signals for multiple function alignment.

Our Bayesian model uses a squared exponential kernel for $\mathcal{K}_g$ with scale factor $\nu = 0.05$ and length scale $\ell = 0.02$, as described in Eq. (4). It employs an exponential kernel for $\mathcal{K}_h$ with $\nu = 0.5$ and $\ell = 10$, and sets the observation noise variance to $\sigma^2 = 0.2^2$. The prior distributions for c_i and e_i are $\text{Gamma}(1, 1)$ and $\mathcal{N}(0, 1000^2)$, respectively.

Fig. 10 presents the alignment results using our proposed Bayesian method. Panel (a) shows the aligned signals alongside their estimated warping functions. In Panel (b), we display the noise-free versions of the aligned functions to better visualize the signal alignment. For comparison, Panel (c) includes the F-R alignment results applied to the noise-free signals, serving as a benchmark. As shown in the figure, our method in Panel (a) produces time warping functions that are nearly identical to the ideal solution shown in Panel (c), which indicates the success of our alignment method even in the presence of additive noise.

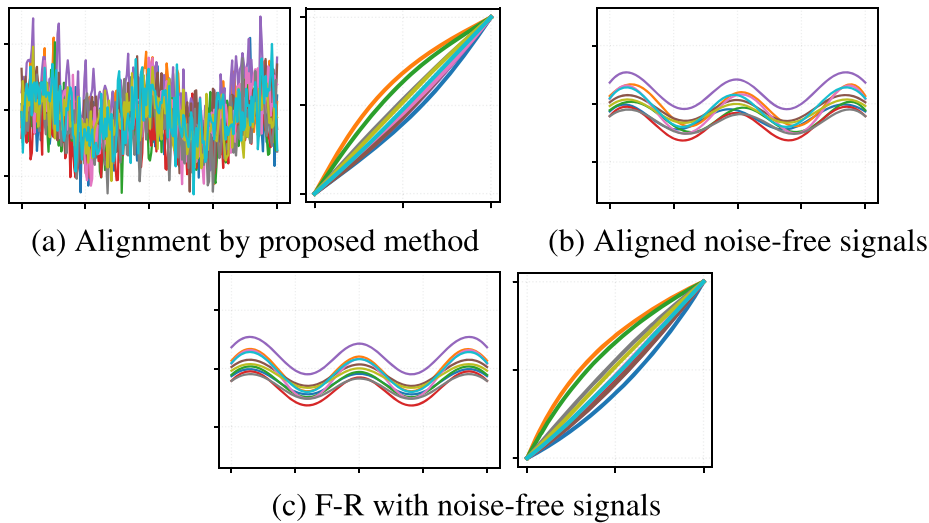


Fig. 10. Alignment results. (a) Aligned y_i 's and posterior mean of warping functions via our proposed method. (b) Corresponding aligned noise-free signals, shown for visual comparison only (not used in estimation). (c) Alignment by F-R of noise-free signals f_i 's with corresponding warping functions.

Table 3

Performance metrics for different alignment methods.

	Least Squares	Pairwise Corr	Sobolev Least Squares
F-R	1.21	0.96	1.95
q -space + SRD	0.97	1.46	0.90
q -space + CLR	0.85	1.63	0.72
f -space + CLR	0.56	2.08	0.22
F-R (noise-free)	0.52	2.21	0.15

Table 4

$\mathbb{L}^2$ distance from the ground-truth g to the estimates of different alignment methods.

	F-R	q -space + SRD	q -space + CLR	f -space + CLR
$\mathbb{L}^2$ Distance	1.00	0.31	0.26	0.08

We also compare the alignment results of our method with those of other approaches, as summarized in Table 3. As expected, our method achieves superior alignment performance across all metrics.

Fig. 11 displays additional inference results. In Panel (a), the log-likelihood trace confirms MCMC convergence. Quantitative diagnostics also support the MCMC convergence. For the pCN updates, the average acceptance rate for h_i 's was approximately 27.5%, and the rate for g was 27.0%. The ESS exceeded 140 per minute for both, indicating efficient sampling. Panels (b) and (c) show the estimated translation and scale parameters, respectively, with ground truth indicated in red. Most importantly, Panel (d) shows the posterior mean of g (dark blue curve) with a 95% credible interval, which closely captures the ground truth signal, highlighted as a red dashed curve. These results demonstrate that the proposed Bayesian model effectively recovers the underlying template function, even in the presence of compositional noise, additive noise, scaling, and translation effects. Table 4 reports the $\mathbb{L}^2$ distances from the ground-truth template for all methods, where our approach achieves the best performance.

Hyperparameter selection for GP priors. The GP kernel hyperparameters are fixed at interpretable values using prior predictive checks, rather than being estimated jointly with the other model parameters. The goal is to choose values that produce reasonable prior variability in the template g and the warping function γ .

- **Template (g):** A squared exponential (SE) kernel is used to encode smooth templates. In the second row of Fig. 12, ℓ_g controls smoothness (larger ℓ_g produces smoother draws), while v_g controls amplitude (larger v_g increases the vertical variation).
- **Warping (h):** An exponential kernel is used for the latent CLR function h . Under the inverse-CLR mapping, (ℓ_h, v_h) influence different geometric features of the induced warping γ . In Fig. 12, ℓ_h mainly controls roughness, while v_h mainly controls deformation magnitude. The two effects can interact, but they are often distinguishable in prior draws. This makes tuning more direct. ℓ_h can be adjusted to target roughness, and v_h can be adjusted to target deformation magnitude.

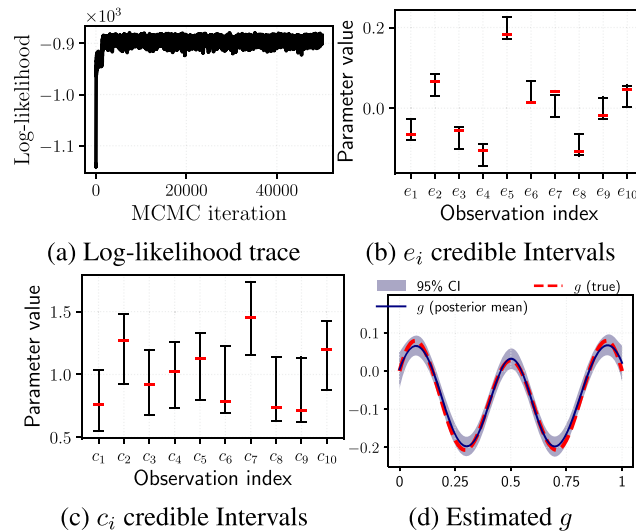


Fig. 11. MCMC additional inference results. (a) Log-likelihood trace over iteration. (b) and (c) Posterior credible intervals for the translation parameters e_i and scale parameters c_i , respectively, with the ground-truth values indicated by red bars. (d) Comparison between the ground-truth underlying function g (dashed red) and the posterior mean estimate of g (solid dark blue). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

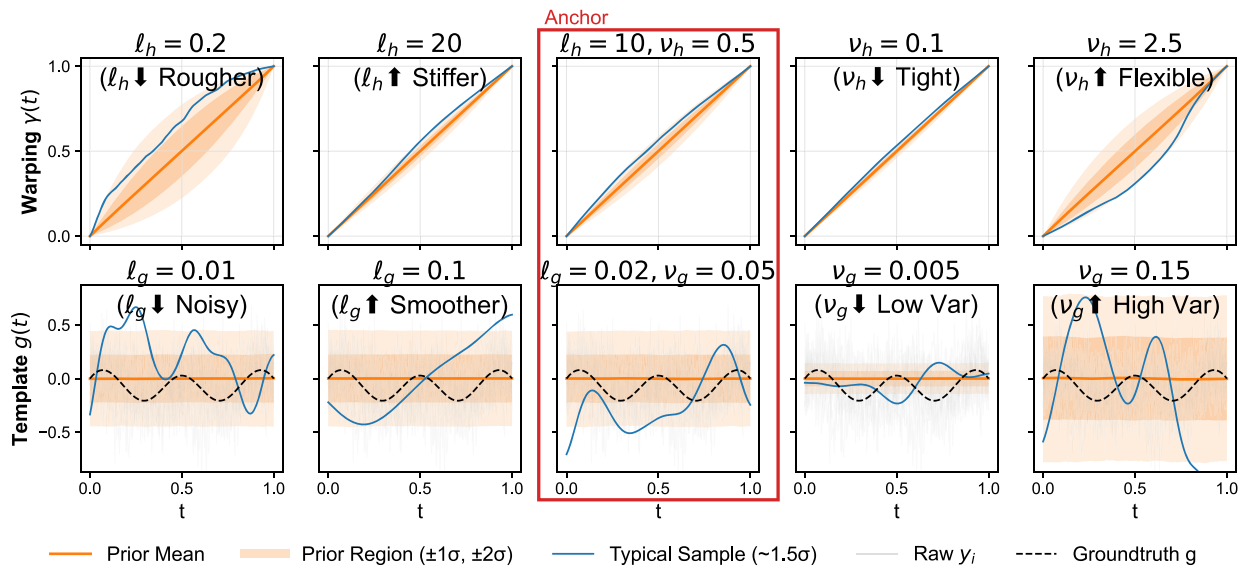


Fig. 12. Prior predictive checks for the example in Section 2.4.2. Top row: prior draws of the warping function $\gamma(t)$. Bottom row: prior draws of the template $g(t)$. The center column shows the selected anchor hyperparameters. The left columns vary the length-scale ℓ with ν fixed, which mainly changes smoothness/roughness. The right columns vary ν with ℓ fixed, which mainly changes amplitude for g and deformation magnitude for γ . Shaded bands show $\pm 1\sigma$ and $\pm 2\sigma$ prior regions around the prior mean.

pCN step-size selection (β). Kernel hyperparameters ℓ and ν set the prior roughness and magnitude of the template g and the induced warping γ . The pCN step size β is tuned separately and scales each proposal. Under the Metropolis–Hastings correction, the choice of β affects proposal size and mixing, but it does not change the target posterior distribution. Randomized or mixture step-size variants of pCN have been used for the same purpose (Cotter et al., 2013; Lu et al., 2017; Ma and Wu, 2025). The value of β is problem dependent. It can be tuned in short preliminary runs to obtain a reasonable acceptance rate. It can also be adjusted during burn-in and then fixed for posterior sampling.

At each iteration, β_h and β_g are drawn from fixed grids. Most values in each grid are small, supporting local exploration. One or two values are large and are reserved for jump proposals. For warping, the grid for β_h has approximately 10 values between 0.03

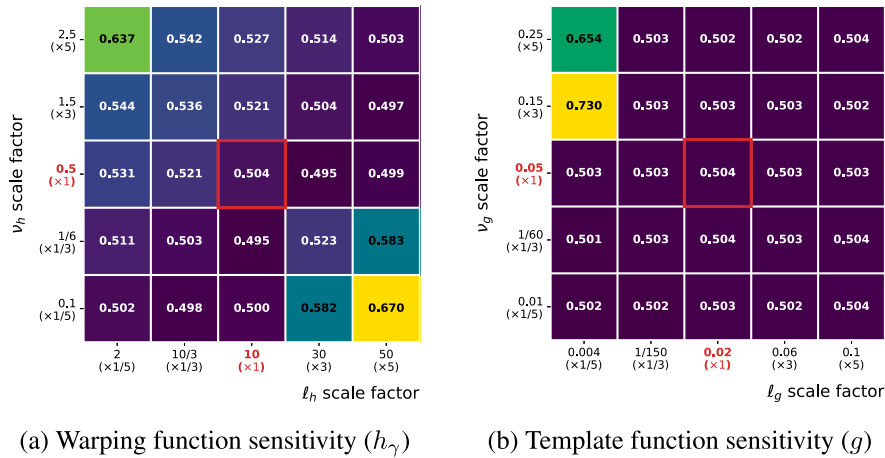


Fig. 13. Sensitivity analysis for the GP hyperparameters. (a) Warping function sensitivity (h_γ). (b) Template function sensitivity (g). The entries show the *least square* metric defined in Section 2.3.2, which measures the reduction in cross-sectional variance after alignment. Red boxes indicate the selected anchor values. Smaller values indicate better alignment.

and 0.45, and a large value is selected with probability approximately 0.1. For the template, the grid for β_g is narrower, from 0.002 to 0.1, to avoid very low acceptance when updating the shared function g .

Sensitivity test for warplings and template functions. Fig. 13 summarizes the sensitivity analysis for the GP hyperparameters. The length-scale and amplitude parameters were varied around the anchor values selected from the prior predictive checks in Fig. 12, with each parameter ranging from 0.2 to 5 times its anchor value. Alignment quality was evaluated using the *least square* metric defined in Section 2.3.2, where smaller values indicate better alignment. The selected anchor values lie in stable regions of the grid. Fig. 13(a) shows larger changes mainly at boundary settings of the warping-prior grid, whereas panel (b) shows that the template-prior grid is stable over most settings, except for a few extreme combinations. Thus, the alignment results are not sensitive to small or moderate changes in the GP hyperparameters around the selected anchors.

Additional MCMC diagnostics, runtime results, and randomized pCN proposal step-size grids for the warping and template updates are provided in Appendix A.

3. Experimental results

We present a more in-depth analysis using both noisy simulated and real-world datasets to evaluate the performance of the proposed Bayesian alignment framework. These experiments are designed to provide empirical evidence of the method's robustness to noise and its flexibility.

3.1. Simulations

In this section, we illustrate two examples to show (1) the flexibility and robustness of the proposed Bayesian alignment method, and (2) the advantage of not relying on the SRVF representation of observed signals in noisy settings. Unlike metric-based optimization methods, the probabilistic nature of our approach allows for multiple plausible alignment solutions. Moreover, by avoiding the SRVF transformation on noisy observations, the model retains robustness to noise while still accurately capturing alignment uncertainty.

3.1.1. Pairwise alignment with different numbers of peaks

We consider a pairwise alignment problem involving signals with differing numbers of peaks and added noise. SRVF-based methods like the F-R method produce a single solution and are sensitive to noise due to their reliance on derivatives. In contrast, our Bayesian approach captures multiple plausible alignments and remains robust in the presence of noise.

The signals are defined with $[t] = [0, 1, 2, \dots, 200]/200$ as follows:

$$y_1([t]) = \exp\left(-\frac{1}{2}\left(\frac{[t] - 0.6}{0.03}\right)^2\right) + \epsilon_2([t]), \text{ and}$$

$$y_2([t]) = \exp\left(-\frac{1}{2}\left(\frac{[t] - 0.3}{0.03}\right)^2\right) + \exp\left(-\frac{1}{2}\left(\frac{[t] - 0.5}{0.03}\right)^2\right) + \exp\left(-\frac{1}{2}\left(\frac{[t] - 0.7}{0.03}\right)^2\right) + \epsilon_1([t]),$$

where $\epsilon_1([t])$ and $\epsilon_2([t])$ are independent noise terms sampled from $\mathcal{N}(0, 0.3^2\mathbf{I})$. In Fig. 14, Panel (a) displays the noisy observations y_1 and y_2 . In the absence of noise, as shown in Panel (b), y_2 contains three distinct peaks, whereas y_1 contains only a single peak.

This setup highlights a key strength of our method: its ability to flexibly explore multiple plausible alignment solutions. The F-R method, being an optimization-based approach, yields a single best alignment by minimizing an SRVF distance functional. However,

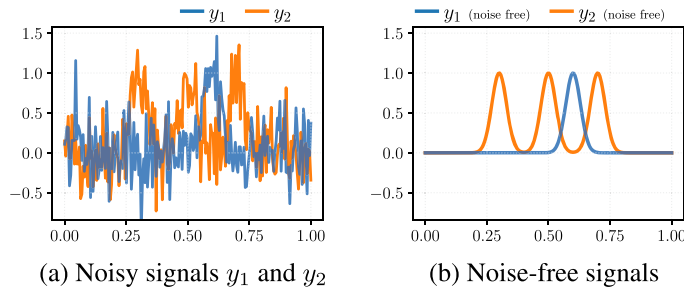


Fig. 14. Pairwise alignment with different numbers of peaks.

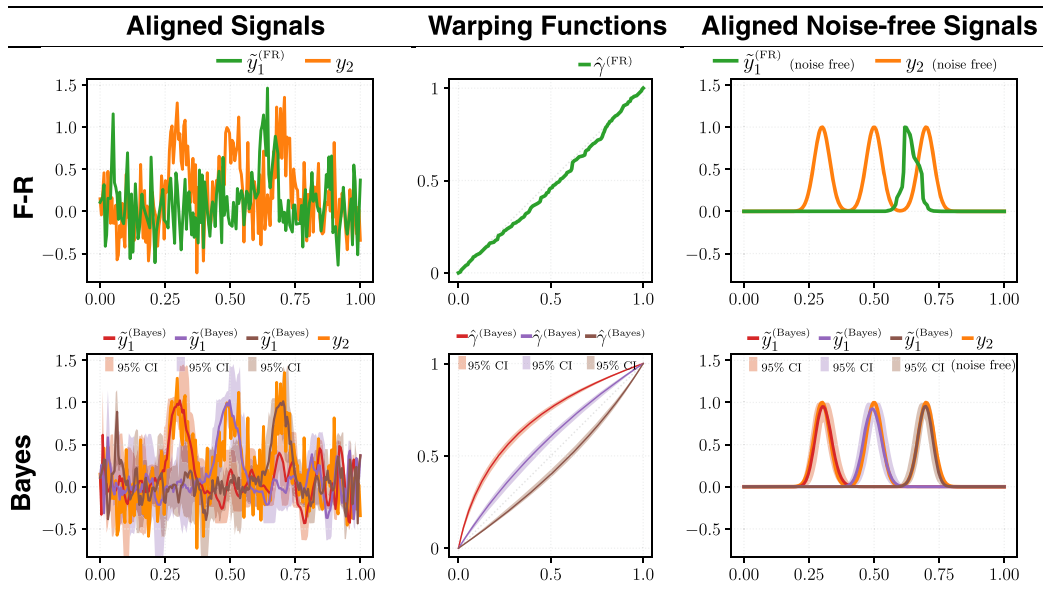


Fig. 15. Alignment results using F-R (top row) and the new Bayesian (bottom row) methods. Left column: Noisy signals y_2 and $\tilde{y}_1$ aligned by each method. Center column: Estimated warping functions for each method. Right column: Corresponding aligned noise-free signals, shown for visual comparison only (not used in estimation).

when multiple plausible alignments exist—as is the case here, where the single peak in y_1 could align with any of the three peaks in y_2 —the F-R approach selects only one and discards the others. In contrast, our Bayesian framework can explore a broad space of warping functions through Z-mixture pCN sampling. This enables our method to explore multiple posterior modes in the space of warping functions, effectively identifying distinct plausible alignments and capturing the associated uncertainty. This is achieved by drawing the mixture weight $\beta_h \in (0, 1)$ from two regions: small values promote local, proximal updates, while large values encourage broader jumps across modes.

Moreover, our method exhibits robustness to additive noise, a critical property that will be further discussed in [Section 3.1.2](#). The flexibility and noise robustness of our method are especially important for real-world signals, which often contain noise and may exhibit ambiguous structures such as multiple features (peaks or transitions) that can reasonably align in more than one way. These structural ambiguities are not well handled by methods that return only a single deterministic alignment.

The alignment results obtained by the F-R and our proposed Bayesian method are summarized in [Fig. 15](#). The top row shows the F-R alignment results, with $\tilde{y}_1^{(FR)}$ in green. The bottom row presents results from our Bayesian method, where three posterior mean functions of $\tilde{y}_1^{(Bayes)}$ are shown in purple, brown, and pink, each accompanied by a 95% credible interval. The left and the middle columns show the aligned signals with additive noise and the warping function, respectively. The right column shows the corresponding alignment applied to the noise-free y_1 using the same warping functions, enabling a clearer view of how each method handles the underlying signal structure.

This illustration clearly shows that our Bayesian approach can recover multiple plausible warping functions and capture uncertainty in the alignment, both of which are unattainable with the F-R method. Moreover, the F-R solution fails to handle the noise effectively, often producing poorly estimated warpings. The underlying reason for this behavior is further explored in the following section.

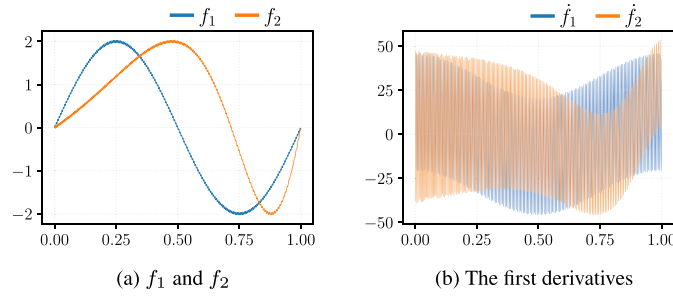


Fig. 16. f_1 and f_2 and their first derivatives, $\dot{f}_1$ and $\dot{f}_2$.

3.1.2. Robustness comparison with the SRVF-based alignment

The SRVF-based alignment methods (Kurtek et al., 2011; Cheng et al., 2016; Lu et al., 2017; Tucker et al., 2021; Ma and Wu, 2025) have been widely adopted due to their elegant geometric formulation and computational efficiency. However, these methods rely on the derivatives of the observed functions, which may make them sensitive to high-frequency noise, even if the noise has low amplitude. As a result, the SRVF representation can be dominated by noise, causing the method to align with the perturbation rather than the underlying signal.

To understand this sensitivity, we examine a fundamental property of the SRVF framework. The SRVF method is based on an important equivalence on phase variability between the space of absolutely continuous functions (referred to as f -space) and the space of their SRVF representations in $\mathbb{L}^2$ (referred to as q -space). Under noise-free settings, for any absolutely continuous function f and its SRVF q , if we define a warped function $\tilde{f} = f \circ \gamma$, then the SRVF of $\tilde{f}$, denoted as $\tilde{q}$, is given analytically by: $\tilde{q}(t) = ((q \circ \gamma) \sqrt{\dot{\gamma}})(t)$. In this case, γ indicates the optimal warping between f and $\tilde{f}$ in the f -space, as well as the optimal one between q and $\tilde{q}$ in the q -space.

However, we point out that when the additive noise is present, the optimal warping in the f -space and q -space may not be the same (or even close). For illustrative purposes, we construct two functions, f_1 and f_2 , in which distinct warping functions are applied to the signal and noise components. Specifically, we define two warping functions $\gamma_1, \gamma_2 \in \Gamma$: γ_1 is applied to the main signal, while γ_2 is applied to the high-frequency noise. The functions are given by:

$$f_1(t) = \underbrace{\sin(2\pi t)}_{\text{main signal}} + \underbrace{\epsilon \sin\left(\frac{t}{\epsilon^2}\right)}_{\text{small, fast oscillation}}, \quad \text{and} \quad f_2(t) = \sin\left(2\pi \underbrace{\gamma_1(t)}_{\text{warping in signal}}\right) + \epsilon \sin\left(\underbrace{\frac{\gamma_2(t)}{\epsilon^2}}_{\text{warping in noise}}\right).$$

In this construction, f_2 exhibits two layers of warping as described. When estimating a single warping function γ to align f_1 with f_2 , the ideal outcome is for γ to recover γ_1 , as it governs the structural alignment of the main signal. In contrast, matching γ_2 would reflect alignment of the noise, which is not desired.

The critical feature of f_1 and f_2 lies in the low-amplitude, high-frequency oscillatory terms scaled by ϵ . Although these perturbations are negligible in the function space, they heavily influence the SRVF representations, which involve the derivatives of the functions:

$$\dot{f}_1(t) = 2\pi \cos(2\pi t) + \frac{1}{\epsilon} \cos\left(\frac{t}{\epsilon^2}\right), \quad \text{and} \quad \dot{f}_2(t) = 2\pi \dot{\gamma}_1(t) \cos(2\pi \gamma_1(t)) + \frac{\dot{\gamma}_2(t)}{\epsilon} \cos\left(\frac{\gamma_2(t)}{\epsilon^2}\right).$$

In both cases, the second term—corresponding to the derivative of the perturbation—is scaled by $1/\epsilon$ and can dominate the overall derivative, even though the perturbation itself is small. As a result, the SRVF-based distance $\|q_2 - (q_1, \gamma)\|_{\mathbb{L}^2}^2$ may be minimized by aligning the noise patterns (via γ_2) rather than the underlying signal (via γ_1). On the other hand, alignment in the original function space using the $\mathbb{L}^2$ metric, $\|f_2 - (f_1 \circ \gamma)\|_{\mathbb{L}^2}^2$, correctly favors warping toward γ_1 , yielding a more meaningful alignment.

To construct this example, we choose the following parameter values:

$$\epsilon = 0.03, \quad \gamma_1(t) = \frac{e^{2t} - 1}{e^2 - 1}, \quad \text{and} \quad \gamma_2(t) = \frac{e^{-0.55t} - 1}{e^{-0.55} - 1}.$$

For numerical accuracy, the time domain is discretized into $T = 501$ equally spaced points: $[t] = \{0, 1/T, 2/T, \dots, 1\}$.

In Fig. 16, Panel (a) shows the resulting functions f_1 and f_2 , each comprising a smooth primary signal and a small high-frequency oscillation. Panel (b) shows their first derivatives, where the high-frequency components are greatly amplified in magnitude, dominating the derivative.

Fig. 17 presents the alignment results using the F-R method (top row) and our proposed Bayesian approach (bottom row). In the F-R results, the aligned signal $\tilde{f}_1^{(\text{FR})}$ and the corresponding warping $\gamma^{(\text{FR})}$ both deviate from the true structure, with $\gamma^{(\text{FR})}$ aligning almost exactly with γ_2 , the warping applied to the noise. In contrast, in our Bayesian framework, the posterior mean of the warping function, $\hat{\gamma}^{(\text{Bayes})}$, closely aligns with the true signal warping, γ_1 , as desired.

This example highlights two critical limitations of the SRVF framework in the presence of additive noise. First, the SRVF relies on function derivatives, making it highly sensitive to high-frequency noise, even when the noise is small in magnitude. Second,

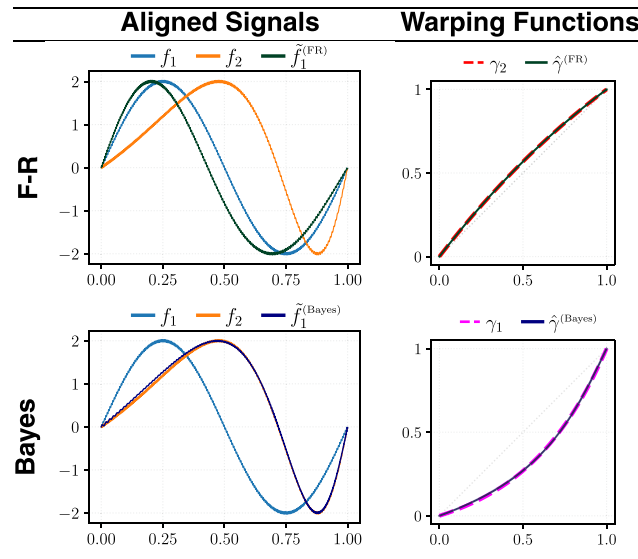


Fig. 17. Alignment comparison between F-R (top row) and Bayesian (bottom row) methods. Left column: Aligned functions $\hat{f}_1^{(FR)}$ (top, green) and $\hat{f}_1^{(Bayes)}$ (bottom, dark blue), overlaid with observed f_1 (light blue) and f_2 (orange). Right column: Estimated warping functions $\hat{\gamma}^{(FR)}$ (top, green) and $\hat{\gamma}^{(Bayes)}$ (bottom, dark blue) with ground truth warping in noise γ_2 (red); and $\hat{\gamma}^{(Bayes)}$ with ground truth warping in signal γ_1 (magenta). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

this sensitivity may lead to undesired results, where the estimated warping possibly aligns with the noise structure rather than the underlying ground-truth signal. In contrast, our Bayesian method, which operates directly in the original function space, remains robust to such perturbations and can result in proper alignment in the data.

3.2. Real data

We demonstrate the practical usages of our Bayesian alignment framework on three real-world datasets. These examples highlight several key advantages of our method: robustness to measurement noise, the ability to incorporate domain-specific prior knowledge, and the capacity to extract representative trends from partially aligned functional observations. For the examples in this section, σ^2 is treated as a fixed parameter that controls the relative weight of the likelihood term in the posterior. We choose σ according to the scale of the observed responses and the goal of the analysis, and report the selected value in each example.

3.2.1. Berkeley's growth rate data

We apply the proposed Bayesian method to the Berkeley female growth dataset, which contains 54 growth rate curves measured quarterly from age 1 to 18. With this dataset, we want to show (1) the robustness to additive noise and (2) the flexibility of our method. We add artificial Gaussian noise to the data to simulate more apparent variability. Specifically, we interpolate the original 69 time points to a finer grid of 100 points and add independent noise terms $\epsilon_i([t]) \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ for $i = 1, \dots, 54$ where $\mathbf{0} \in \mathbb{R}^{100}$ and $\mathbf{I} \in \mathbb{R}^{100 \times 100}$. For the Bayesian fit in this example, we fixed $\sigma^2 = 0.2^2$. This likelihood scale emphasizes the main growth-spurt features, while the warping prior discourages alignment to local noisy features.

1. Alignment with additional high noise-level:

To assess robustness under increased noise, we compare alignment results from our proposed method and the F-R approach, both with and without additional additive noise. Fig. 18 shows the original dataset in Panel (a) and the noise-added version in Panel (b). As expected, the functions in Panel (b) appear significantly choppier due to the added noise.

Fig. 19 illustrates the alignment performance of our proposed method on real growth data. Panel (a) shows that our method successfully aligns the signals even in the presence of additional additive noise, revealing two prominent peaks around ages 5 and 12. For a more precise comparison, Panel (b) presents the noise-free version of these aligned signals. These can be directly compared with Panel (c), which shows the results of the F-R alignment applied to the original (noise-free) signals.

Panel (d) contrasts the estimated warping functions g produced by our method (solid blue) and the F-R method (dashed black) on the noise-free data. Notably, in the age range of 1 to 10, the F-R approach tends to over-align by matching all observable peaks, potentially including noisy fluctuations in the original signals. In contrast, our method avoids such over-alignment, albeit at the cost of merging multiple peaks into a single, broader, and less distinct one. Admittedly, in the absence of known ground truth alignments, it remains an open question which method yields a more faithful representation for this particular age range. Nevertheless, both methods consistently detect the final growth spurt around age 12, suggesting that alignment efforts may

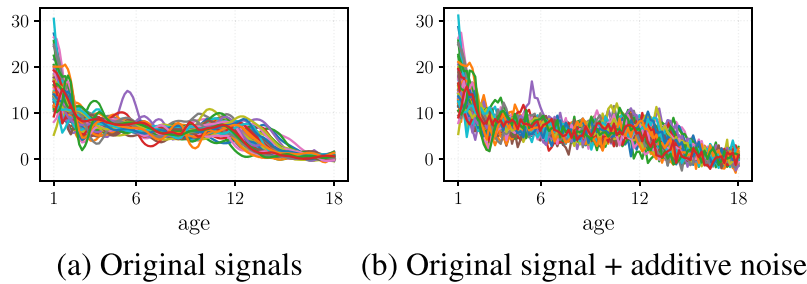


Fig. 18. Berkeley growth rate data: (a) original signals; (b) signals with added noise.

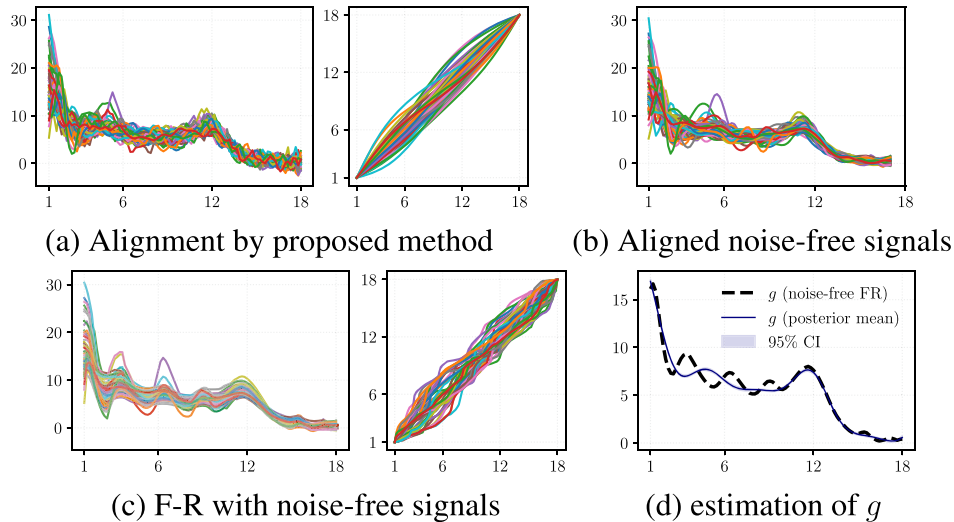


Fig. 19. Alignment results and comparison. (a) Alignment results via the proposed method: (left) aligned noisy functions f_i 's and (right) corresponding warping functions. (b) Additional noise-free aligned functions using the proposed method, shown for comparison. (c) Alignment results using the F-R method with the original noise-free functions. (d) Posterior mean of g from our method (solid blue) compared with the F-R estimate under noise-free data (dashed black). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Table 5

Performance metrics for different alignment methods.

	Least Squares	Pairwise Corr	Sobolev Least Squares
F-R	1.24	1.00	1.23
q -space + SRD	0.93	1.02	0.90
q -space + CLR	1.03	1.04	1.03
f -space + CLR	0.70	1.07	0.75
F-R (noise-free)	0.84	1.08	0.40

reasonably prioritize this last prominent peak. This can be further refined by incorporating prior information on the warping functions, as will be illustrated in the examples that follow.

In Table 5, we also compare the alignment performance of the alternative methods, including F-R and SRVF-based Bayesian approaches, under the presence of additional additive noise. Unlike the alternatives, which tend to struggle to align noisy curves effectively, our approach accurately recovers the key growth patterns and produces a smoother, biologically meaningful underlying signal g . Notably, our method outperforms all others across the three evaluation criteria, achieving consistently superior alignment accuracy.

2. Incorporating warping prior:

In many applications, researchers hold strong prior beliefs about when meaningful variation in a signal occurs. For instance, one may believe that significant growth spurts occur only after age 9, and that growth before this age is of less scientific interest. In this case, we can encode such prior knowledge by manipulating the hyperparameter $\mathcal{K}_h$, as illustrated in Fig. 20(a).

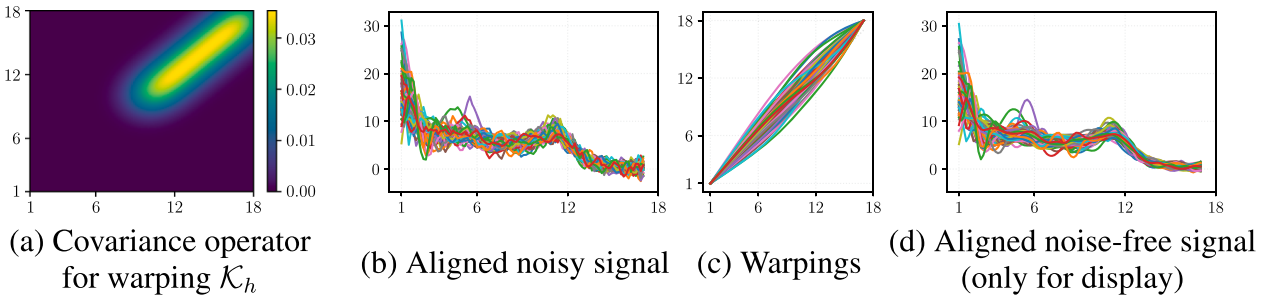


Fig. 20. Prior and alignment results. (a) Prior covariance structure $\mathcal{K}_h$ restricting warpings to occur after age ten. (b) Aligned noisy functions via our proposed method. (c) Corresponding estimated warping functions. (d) Noise-free representation of (b), shown for comparison only (note: noise-free data were not used during estimation).

To impose biologically informed constraints on warping functions, we define a structured Gaussian process prior using the covariance kernel $\mathcal{K}(s, t) = r(t) \delta(s - t)$, where $\delta(\cdot)$ is the Dirac delta function, and $r(t) = \begin{cases} v_h, & \text{if } t \in [10, 18], \\ 0, & \text{otherwise,} \end{cases}$ with $v_h = 1$. This construction encodes the prior belief that warping variability is unnecessary during early childhood (ages 1–9). However, the final growth spurt during adolescence is marked by substantial individual variation in timing and pace, justifying increased flexibility in alignment. The resulting diagonal structure is then smoothed using a two-dimensional Gaussian convolution with bandwidth 0.08, producing a continuous covariance function $\mathcal{K}_h(s, t)$ that diffuses influence across neighboring ages. A small diagonal component is added to ensure positive definiteness. The final covariance operator $\mathcal{K}_h$ is thus both smooth and structured: it enforces nearly uniform warping in early ages while allowing flexibility during adolescence. The illustration of this specific $\mathcal{K}_h$ is shown in Fig. 20(a).

A similar modeling framework without the additive noise setting was previously studied in Ma and Wu (2025). Here, we extend that setup by introducing additional noise to the data and demonstrate that our method continues to perform well under these more challenging conditions. Fig. 20(b) illustrates the resulting alignment. As expected, the time-warping function remains nearly rigid in the age range 1 to 9, while active alignment occurs between ages 9 and 18, where individual variation is most pronounced.

In summary, this example highlights that incorporating strong prior information remains effective even under high-noise conditions. In contrast, the F-R approach, being non-Bayesian, lacks a prior and does not include a systematic modeling framework for warping functions. Furthermore, other Bayesian approaches that utilize the SRVF representation of warping functions cannot explicitly model warping functions, as the nonlinear exponential mapping distorts variability. In comparison, the CLR representation preserves this variability, allowing for more direct, flexible, and interpretable modeling.

3.2.2. COVID-19 death rate data (25 European countries during 2020 - 2022)

This example illustrates how structural constraints in our Bayesian framework prevent implausible warpings and preserve meaningful temporal relationships, avoiding alignments between distant events that could distort epidemiological interpretation. The dataset reports the 7-day rolling average of COVID-19-related deaths per one million population (Mathieu et al., 2020). The raw signals are presented in Fig. 21(a), where two major waves are evident: the first from September 2020 to May 2021, and the second from July 2021 to February 2022.

In this application, we emphasize the importance of avoiding overly aggressive warping during the alignment process. Specifically, we aim to discourage scenarios where two peaks that are far apart in time in the raw signals are aligned closely in the warped signals. Such cases are biologically and epidemiologically implausible, as huge temporal gaps diminish any meaningful correlation. For instance, aligning peaks that are a year apart may obscure their underlying causes and violate assumptions of local temporal dependence. We point out that aggressive alignment methods (e.g., F-R) may allow significant temporal shifts to minimize an alignment cost function, without accounting for temporal plausibility.

For this analysis, we fixed $\sigma^2 = 1$. To address the need for targeted temporal flexibility, we impose soft constraints on the prior covariance operator $\mathcal{K}_h$ for the warping functions. Specifically, we construct a structured Gaussian process prior using the kernel of the form, $\mathcal{K}(s, t) = r(t) \delta(s - t)$, where $\delta(\cdot)$ denotes the Dirac delta function, and

$$r(t) = \begin{cases} v_h, & \text{if } t \in [\text{Sep 2020, May 2021}] \cup [\text{Jul 2021, Feb 2022}], \\ 0, & \text{otherwise,} \end{cases}$$

with $v_h = 1$. This reflects the prior belief that alignment flexibility should be concentrated within two epidemiologically meaningful periods of the COVID-19 pandemic. This covariance operator is then smoothed using a two-dimensional Gaussian convolution with bandwidth 0.08, resulting in a continuous covariance structure $\mathcal{K}_h(s, t)$ that diffuses the influence of $r(t)$ across neighboring time points.

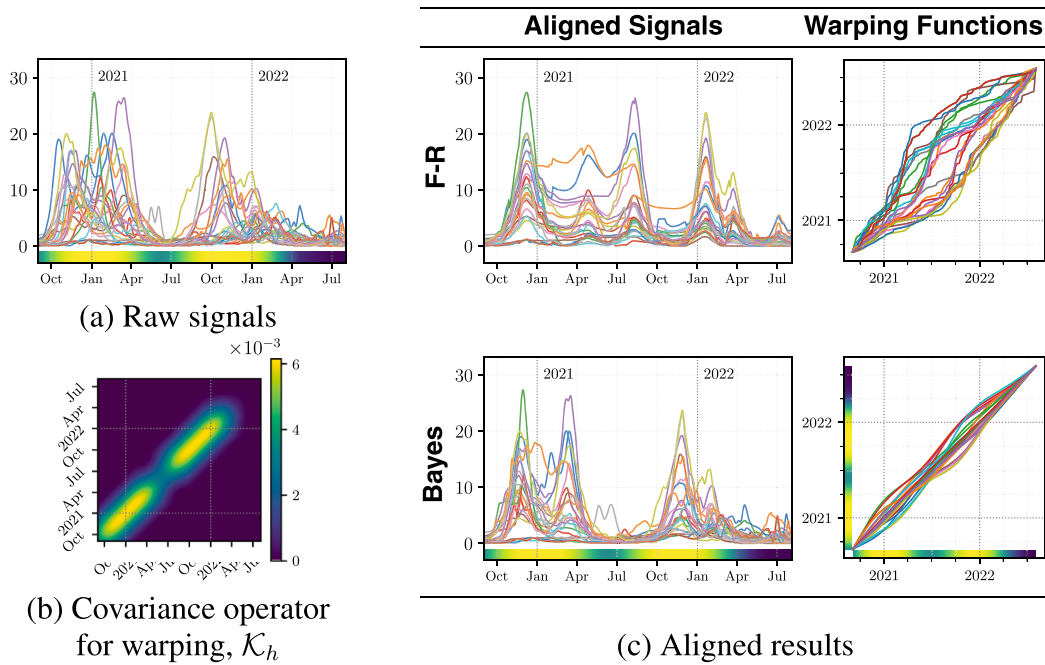


Fig. 21. Comparison of raw and aligned signals. (a) Raw COVID-19 death rate functions. (b) Prior covariance operator $\mathcal{K}_h$, which imposes shape restrictions on the warping functions. (c) Alignment results: F-R method (top row) and Bayesian method (bottom row), showing aligned functions (left) and estimated warping functions (right).

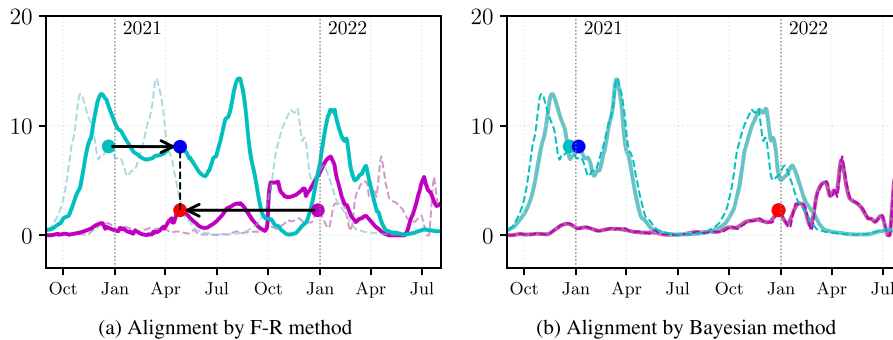


Fig. 22. Alignment comparison with two signals. (a) Alignment using the F-R method. Dotted curves represent raw signals; solid curves show aligned signals. The magenta point is warped to the red point, and the cyan point is warped to the blue point. (b) Alignment using our proposed Bayesian method with the identical layout as (a). The near-perfect overlap of the magenta dotted and solid curves indicates negligible warping. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

The resulting operator $\mathcal{K}_h$ is both smooth and structured: it encourages non-linear time-warping where meaningful dynamics are expected and enforces rigidity outside these periods. The form of $\mathcal{K}_h$ is visualized in Fig. 21(b). These regions guide the Bayesian alignment process by indicating where temporal deformation is permitted to adapt most actively.

Panel (c) presents the alignment results for F-R (top) and our proposed approach (bottom). In the F-R results, over-alignment is clearly visible, and the corresponding warping functions are highly nonlinear, showing large deviations. In contrast, our proposed method reduces this problem by discouraging extreme warpings, leading to better alignment—most adjustments occur within each major epidemic wave. To further illustrate this, Fig. 22(a) presents two examples of extreme warping by the F-R method. In one case (magenta curve), a peak located originally in January 2022 (magenta dot) is shifted back to April 2021 (red dot), spanning nearly seven months, making it difficult to justify. In another (cyan curve), a peak from January 2021 (cyan dot) is warped to align with April 2021 (blue dot), bridging two temporally distant features (cyan and magenta dots) that have a one-year gap. As a result, the overall aligned signals in the top row of Fig. 21 exhibit noticeable distortions compared to the raw signals.

Our Bayesian method yields more plausible alignment results. As shown in Fig. 22(b), no extreme or implausible alignments are observed. Similarly, the overall alignment results in the bottom row of Fig. 21(c) demonstrate that the signals remain close to their original timing, while still successfully aligning nearby peaks across curves. This highlights the benefit of incorporating prior

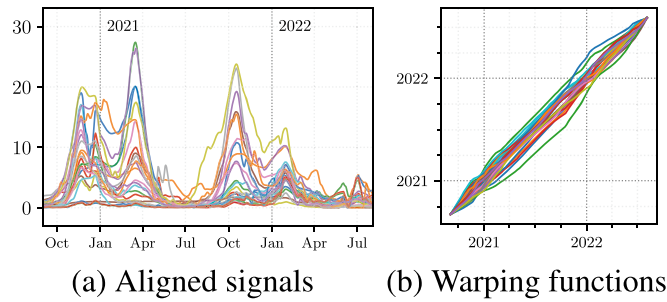


Fig. 23. Alignment results by penalized F-R.

structural information, specifically, guiding when and where warping should be more or less active. Such informed constraints enable more realistic alignment, particularly in datasets with strong temporal structure, such as those related to pandemic wave patterns.

The over-alignment issue can also be mitigated using the penalized F-R method. Fig. 23 presents its results. Panel (a) shows that penalized F-R achieves alignment largely within each major wave, producing results comparable to our Bayesian method in terms of the aligned signals. The main distinction appears in panel (b): the estimated warping functions. Penalized F-R applies a general penalty that simply pulls the warpings toward the identity γ_{id} , without encouraging any particular structural form. In contrast, our Bayesian framework allows explicit control over the prior structure, enabling warping patterns to reflect meaningful domain knowledge. While neither method is universally superior, the Bayesian approach may be more suitable in applications where strong prior information about temporal structure is available.

3.2.3. COVID-19 hospitalization rate data (12 US states during 2020 - 2025)

This example demonstrates how our Bayesian framework estimates a representative signal from structurally similar signals. Unlike conventional averaging, which can be unreliable under noise, our method jointly smooths and aligns the data, capturing shared trends while accounting for timing variation and uncertainty.

The dataset presents the weekly COVID-19 hospitalization rates from March 2020 to April 2025 for 12 U.S. states — California, Colorado, Connecticut, Georgia, Maryland, Michigan, Minnesota, New Mexico, New York, Oregon, Tennessee, and Utah — based on data collected by the Centers for Disease Control and Prevention (CDC). Hospitalization rates are defined as the number of residents in a surveillance area hospitalized with laboratory-confirmed SARS-CoV-2 infection, divided by the total population estimate for that area, and expressed per 100,000 people (Control, 2025).

One key advantage of our method lies in its ability to estimate a representative signal across similar curves. In this case, since the data reflect domestic hospitalization trends across multiple states, the overall shape and seasonal pattern of the curves are relatively similar. As shown in Fig. 24(a), the raw data reveal unaligned trends with a shared structure, and the red curve represents their cross-sectional mean.

Suppose we are interested in analyzing the seasonality and long-term trends of COVID-19 hospitalization rates. In such a setting, accurately estimating a representative signal is especially valuable, as it allows us to capture consistent patterns across states while accounting for state-level variability. For this analysis, we fixed $\sigma^2 = 5^2$. This choice allows softer registration, discouraging alignment to short-term local features and helping the estimated template capture the shared seasonal and long-term trend.

To address seasonal effects, we incorporate two sources of prior information: one for the warping functions, h_{γ_i} , and another for the underlying seasonal trend signal, g . As in the previous example, we specify the covariance structure of the warping functions h_{γ_i} as $\mathcal{K}_h(s, t) = 5 \delta(s - t)$. To prevent extreme alignment shifts exceeding a six-month range, $\mathcal{K}_h$ is then smoothed using a two-dimensional Gaussian convolution with a bandwidth of half a year, as illustrated in Fig. 24(b).

For the prior on g , which governs the smoothness of the seasonal trend signal, we adopt a squared exponential kernel: $\mathcal{K}_g(s, t) = 10 \exp\left(-\frac{(s-t)^2}{0.005}\right)$. Given that the data spans a five-year period, this choice implies an expected number of approximately 5 peaks under the prior.

The alignment results of our method are presented in Fig. 24(c), second row. As shown in the estimated warping functions, no peak alignments with extreme temporal gaps are observed. The estimated representative signal g (shown in dark blue) consistently identifies a single peak each year, occurring around January, effectively capturing the underlying seasonal trend. In contrast, the F-R method tends to over-align non-significant peaks, as shown in the warping function plot from 2020 to 2022 in the first row, where unnecessary temporal stretching is observed.

Such over-alignment can adversely affect signal estimation. A conventional approach to signal estimation is to take the cross-sectional mean of the aligned signals, as in the F-R method in the first row of Fig. 24(c). While this may be reasonable in a noise-free context, it becomes unreliable when the signals are noisy, as the resulting estimate inherits the irregularities and noise present in the data. In contrast, our framework provides a joint, model-based smoothing approach. Unlike traditional smoothers that act independently on each curve, our method estimates the underlying signal g by accounting for both phase variation (via the warping functions γ_i) and shared trend structure.

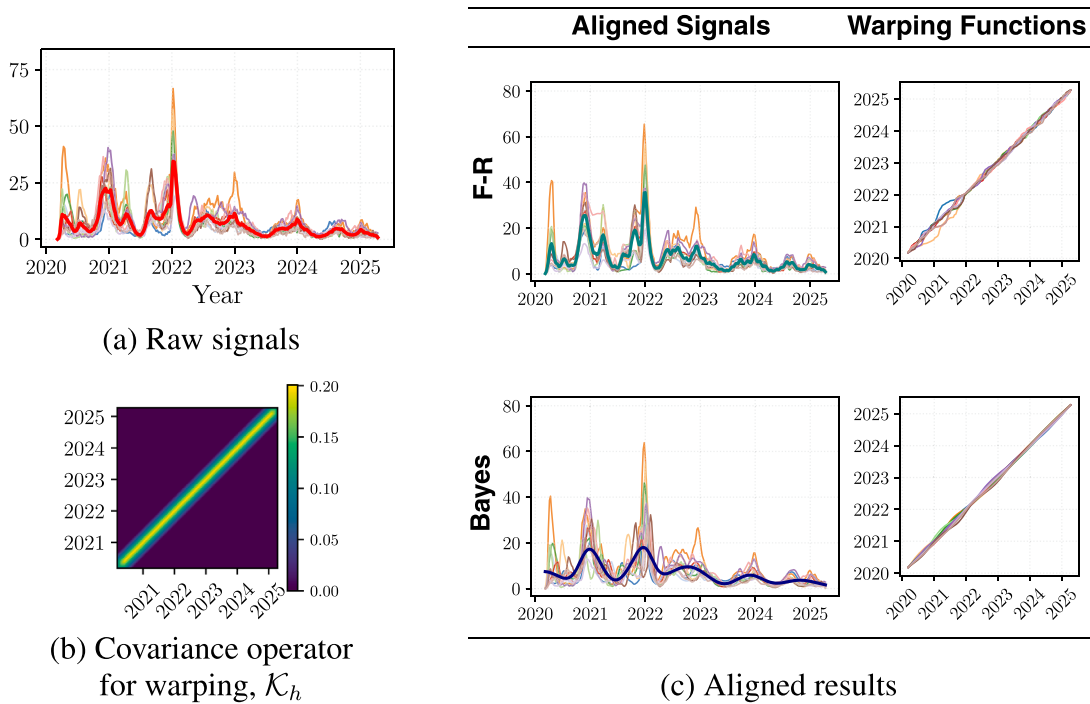


Fig. 24. Comparison of raw and aligned signals. (a) Raw COVID-19 hospitalization rate functions, with the cross-sectional mean (bold red). (b) Prior covariance operator $\mathcal{K}_h$ for warping. (c) Alignment results: F-R method (top row) and Bayesian method (bottom row), showing aligned functions (left) with cross-sectional mean in bold cyan and the posterior mean of g in bold blue, and estimated warping functions (right). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

4. Discussions

Our Bayesian framework offers both flexibility in modeling and robustness to additive noise. In this study, we address not only functional registration but also signal estimation under noisy observations, broadening the scope beyond alignment alone. Function alignment under additive noise has been studied for decades (Kneip and Gasser, 1992; Ramsay and Li, 1998; Gervini and Gasser, 2004; James, 2007); nevertheless, to the best of our knowledge, this study is the first to offer a systematic and principled approach to this problem with a clear and comprehensive demonstration. Remarkably, even in the presence of additive noise, our method achieves performance comparable to the F-R method (Kurtek et al., 2011) applied to noise-free data. Moreover, when evaluated under noisy conditions, our method not only matches but outperforms both the Fisher-Rao approach and recent SRVF-based Bayesian models (Lu et al., 2017; Ma and Wu, 2025). Notably, despite being tailored for noisy data, our framework remains highly effective in the absence of noise, demonstrating its broad applicability and robustness across varying real-world data conditions.

In contrast to our probabilistic modeling framework, penalty-based optimization methods such as those using second-derivative (Ramsay and Li, 1998; Kneip and Ramsay, 2008), first-derivative (Srivastava and Klassen, 2016), or CLR prior (Ma and Wu, 2025) are not model-based. These methods primarily aim to control the variability of warping functions by enforcing closeness to the identity warping γ_{id} . As a result, they typically yield a single optimal warping and lack the ability to capture alignment uncertainty. Our model, by contrast, provides a full posterior distribution over warping functions, enabling richer representation and quantification of uncertainty. This is particularly advantageous in applications where multiple plausible alignments may exist.

Admittedly, SRVF's isometry property under warping (i.e., $\|q\|_{\mathbb{L}^2}^2 = \|(q, \gamma)\|_{\mathbb{L}^2}^2$ for any $\gamma \in \Gamma$ and $q \in \mathbb{L}^2([0, 1])$) equips the F-R method, a metric-based framework, with several significant mathematical properties:

1. Although extensively studied, so far, there is no standard definition of optimal alignment between two functions. The F-R metric provides a rigorous foundation for addressing this issue. It also provides a powerful and convenient framework for studying asymptotics.
2. This metric leads to several desirable properties in alignment. For example, the symmetry can be achieved between two functions—if γ is optimal from f to g , then γ^{-1} is optimal from g to f . This symmetry enhances the interpretability of the alignment process. Additionally, the notion of a mean warping function can be defined with the metric, which is essential in aligning multiple functions.
3. The optimal warping under this metric is invariant to scaling and translation. This is important as it allows us to focus primarily on the *shape* of functions, rather than their size or location.

The F-R framework, however, fundamentally allows a single point estimate for the warping function. In contrast, our approach is grounded in probabilistic modeling, combining a likelihood with a prior over warpings, and performing inference via modern function-space MCMC algorithms. This distinction leads to several practical advantages.

1. **Flexibility:** It accommodates prior information and yields posterior uncertainty quantification, allowing for more interpretable and robust alignment, especially in scenarios with multiple plausible warpings (see [Section 3.1.1](#)).
2. **Robustness to additive noise:** Our method is significantly more robust to additive noise. The F-R method relies on the SRVF representation, which assumes absolute continuity of observed functions, a condition often violated under additive noise. Even with smooth noise, SRVF-based alignment can be unstable (see [Section 3.1.2](#)).
3. **Statistical modeling of warping variability:** Our use of the CLR transformation allows a warping function represented within an $\mathbb{L}^2$ subspace via an isometric isomorphism. This enables more accurate statistical modeling compared to the curved SRD manifold used in the SRVF framework. Such structured modeling may not be feasible with the penalized F-R, where the penalty typically enforces only closeness to the identity warping, γ_{id} .
4. **Computational efficiency:** Our method is more efficient for high-resolution data. While the F-R method relies on dynamic programming with quadratic complexity, our MCMC-based approach scales linearly with the number of time points. This provides substantial efficiency gains for high-resolution data. However, we acknowledge that MCMC methods may involve slower convergence or inefficiencies in the sampling process, depending on the complexity of the posterior distribution.

Compared to other Bayesian methods ([Lu et al., 2017](#); [Tucker et al., 2021](#); [Ma and Wu, 2025](#)), our approach also offers notable improvements:

1. **Robustness to additive noise:** Our model is substantially more robust to additive noise. By working directly in the f -space (see [Section 3.1.2](#)), we avoid the amplification of numerical errors that can arise in the q -space. Furthermore, the pinching effect often seen with $\mathbb{L}^2$ alignment is mitigated by imposing a CLR prior, which enforces bounded derivatives on the warping functions. This prior promotes well-behaved warpings even in the presence of noisy observations.
2. **Simpler and more stable inference:** Our inference procedure is both simpler and more stable. By operating in the f -space, we sidestep the structural complexity and instability associated with the SRVF representation. Notably, our method enables direct estimation of the underlying signal from posterior samples. In contrast, SRVF-based models require recovering the signal via the inverse SRVF transform, which involves numerical integration. This integration step tends to accumulate errors over time, resulting in increased variability during the alignment process.

Beyond the registration methods discussed above, recent work in shape analysis has also considered uncertainty quantification. [Luo and Strait \(2024\)](#) proposed a multiple-output Gaussian process framework for modeling collections of closed planar curves and quantifying uncertainty in shape-analysis tasks. Their focus is shape-level uncertainty for closed curves, whereas ours is uncertainty in subject-specific warpings.

Our future directions aim to expand the flexibility and applicability of the proposed Bayesian framework. Relatedly, [Bharath and Kurtek \(2020\)](#) proposed flexible priors for warping functions. In our framework, landmark pairs could be incorporated as optional hard or soft constraints; we leave this as future work. Another future direction concerns the prior construction. The current implementation places a Gaussian prior directly on the function space to satisfy the requirements of the pCN algorithm; several extensions are worth exploring. These include: (1) applying the method to additional real-world datasets where additive noise is present; (2) systematically comparing the practical performance of different MCMC algorithms, such as ∞ -HMC and Z-mixture pCN; and (3) developing non-Gaussian prior models for the warping functions, potentially via finite-dimensional basis expansions that enable more expressive stochastic process modeling.

Appendix A. MCMC diagnostics and computational details

Additional MCMC diagnostics, runtime results, and randomized pCN proposal step-size grids are provided in [Appendix A. Table A.1](#) reports acceptance rates and effective sample size (ESS) per minute for the function-valued MCMC updates. For Algorithms 7, 16, and 19, the diagnostics for h_γ summarize the subject-specific warping updates.

[Fig. A.1](#) summarizes the wall-clock runtime for [Algorithm 19](#) under different grid sizes T and sample sizes N . Runtime was measured in a standard Google Colab environment using the median of five chains, each run for 10,000 iterations. Empirically, the runtime increases approximately as $O(T^{1.2})$ with the grid size and approximately linearly with the number of functions N .

Table A.1

Quantitative diagnostics for the proposed MCMC algorithms. [Algorithm 7](#) has no template update, so diagnostics for g are not applicable.

Algorithm	Acceptance rate (%)		ESS per minute	
	h_γ	g	h_γ	g
Alg 1 (Pairwise)	9.1%	–	380.7	–
Alg 2 (Multiple)	22.8%	23.8%	111.9	114.1
Alg 3 (Signal Est.)	27.5%	27.0%	147.1	143.9

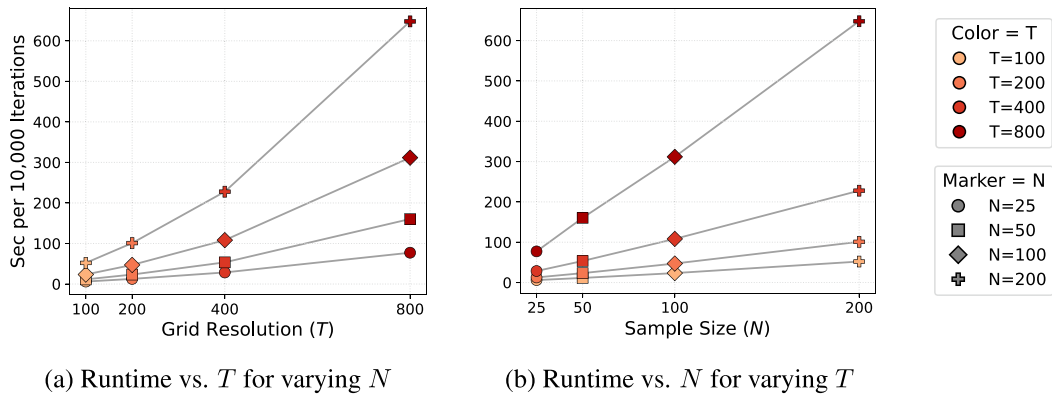


Fig. A.1. Median wall-clock runtimes for Algorithm 3 over five chains with 10,000 iterations each. Runtime increases approximately as $O(T^{1.2})$ with the grid size T and linearly with the number of functions N .

For the randomized pCN updates of the CLR-transformed warplings h_i and the template function g , the proposal step sizes β_h and β_g are sampled from fixed grids. These step sizes affect proposal size, acceptance rate, and mixing, but they do not change the target posterior distribution under the Metropolis–Hastings correction (Cotter et al., 2013; Lu et al., 2017; Ma and Wu, 2025). The grids were chosen using short pilot runs to avoid very low acceptance rates while keeping the chains moving.

In the reported experiments, β_h ranges roughly from 0.03 to 0.45. This wider grid allows both small local moves and occasional larger moves for the warping functions. For the template function, β_g ranges roughly from 0.002 to 0.1. This grid is narrower because g is shared across observations, so large template proposals are more likely to reduce the likelihood sharply and be rejected.

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